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GAN/gan.py

```python
import numpy as np

import torch
import torch.nn as nn
import torchvision.transforms as T
import torch.optim as optim
from torch import Tensor
from torch.utils.data import DataLoader

NOISE_DIM = 96

device = torch.device('cuda') if torch.cuda.is_available() else (torch.device('mps') if torch.backends.mps.is_available() else torch.device('cpu'))


def sample_noise(batch_size: int, dim: int, seed: int | None = None) -> Tensor:
    """
    Generate a PyTorch Tensor of uniform random noise.

    Input:
    - batch_size: Integer giving the batch size of noise to generate.
    - dim: Integer giving the dimension of noise to generate.

    Output:
    - A PyTorch Tensor of shape (batch_size, dim) containing uniform
      random noise in the range (-1, 1).
    """
    if seed is not None:
        torch.manual_seed(seed)
    return torch.rand(batch_size, dim) * 2 - 1


class Discriminator(nn.Module):
    def __init__(self):
        super().__init__()

        ############################################################################
        # TODO: Implement architecture                                             #
        #                                                                          #
        # HINT: nn.Sequential might be helpful. You'll start by calling nn.Flatten()#
        ############################################################################
        # *****START OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****

        self.model = nn.Sequential(
            nn.Flatten(),
            nn.Linear(784, 256),
            nn.LeakyReLU(0.01),
            nn.Linear(256, 256),
```

```
            nn.LeakyReLU(0.01),
            nn.Linear(256, 1),
        )

        # *****END OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
        ############################################################################
        #                               END OF YOUR CODE                           #
        ############################################################################

    def forward(self, x):
        return self.model(x)


class Generator(nn.Module):
    def __init__(self, noise_dim: int = NOISE_DIM):
        super().__init__()
        self.noise_dim = noise_dim

        ############################################################################
        # TODO: Implement architecture                                             #
        #                                                                          #
        # HINT: nn.Sequential might be helpful.                                    #
        ############################################################################
        # *****START OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****

        self.model = nn.Sequential(
            nn.Linear(self.noise_dim, 1024),
            nn.ReLU(),
            nn.Linear(1024, 1024),
            nn.ReLU(),
            nn.Linear(1024, 784),
            nn.Tanh(),
        )

        # *****END OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
        ############################################################################
        #                               END OF YOUR CODE                           #
        ############################################################################

    def forward(self, x):
        return self.model(x)


def bce_loss(input: Tensor, target: Tensor) -> Tensor:
    """
    Numerically stable version of the binary cross-entropy loss function in PyTorch.

    Inputs:
    - input: PyTorch Tensor of shape (N, 1) giving scores.
    - target: PyTorch Tensor of shape (N, 1) containing 0 and 1 giving targets.
```

```
          uses torch.float32, not an integer type.

    Returns:
    - A PyTorch Tensor containing the mean BCE loss over the minibatch of input data.
    """
    bce = nn.BCEWithLogitsLoss()
    return bce(input, target)


def discriminator_loss(logits_real: Tensor, logits_fake: Tensor) -> Tensor:
    """
    Computes the discriminator loss described above.

    Inputs:
    - logits_real: PyTorch Tensor of shape (N, 1) giving scores for the real data.
    - logits_fake: PyTorch Tensor of shape (N, 1) giving scores for the fake data.

    Returns:
    - loss: PyTorch Tensor containing (scalar) the loss for the discriminator.

    HINT: Use torch.ones_like / torch.zeros_like to create label tensors that
    automatically match the device and dtype of the input.
    """
    loss = None
    # *****START OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****

    real_labels = torch.ones_like(logits_real)
    fake_labels = torch.zeros_like(logits_fake)
    loss = bce_loss(logits_real, real_labels) + bce_loss(logits_fake, fake_labels)

    # *****END OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
    return loss


def generator_loss(logits_fake: Tensor) -> Tensor:
    """
    Computes the generator loss described above.

    Inputs:
    - logits_fake: PyTorch Tensor of shape (N, 1) giving scores for the fake data.

    Returns:
    - loss: PyTorch Tensor containing the (scalar) loss for the generator.

    HINT: Use torch.ones_like to create label tensors that automatically match
    the device and dtype of the input.
    """
    loss = None
    # *****START OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****

```

```
    fake_labels = torch.ones_like(logits_fake)
    loss = bce_loss(logits_fake, fake_labels)

    # *****END OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
    return loss


def get_optimizer(model: nn.Module) -> torch.optim.Adam:
    """
    Construct and return an Adam optimizer for the model with learning rate 1e-3,
    beta1=0.5, and beta2=0.999.
    """
    return optim.Adam(model.parameters(), lr=1e-3, betas=(0.5, 0.999))


def ls_discriminator_loss(scores_real: Tensor, scores_fake: Tensor) -> Tensor:
    """
    Compute the Least-Squares GAN loss for the discriminator.

    Inputs:
    - scores_real: PyTorch Tensor of shape (N, 1) giving scores for the real data.
    - scores_fake: PyTorch Tensor of shape (N, 1) giving scores for the fake data.

    Outputs:
    - loss: A PyTorch Tensor containing the loss.
    """
    loss = None
    # *****START OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****

    loss = 0.5*torch.mean((scores_real-1)**2)+0.5*torch.mean((scores_fake)**2)

    # *****END OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
    return loss


def ls_generator_loss(scores_fake: Tensor) -> Tensor:
    """
    Computes the Least-Squares GAN loss for the generator.

    Inputs:
    - scores_fake: PyTorch Tensor of shape (N, 1) giving scores for the fake data.

    Outputs:
    - loss: A PyTorch Tensor containing the loss.
    """
    loss = None
    # *****START OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****

    loss = 0.5*torch.mean((scores_fake-1)**2)

```

```
    # *****END OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
    return loss


class DCDiscriminator(nn.Module):
    def __init__(self):
        super().__init__()

        ############################################################################
        # TODO: Implement architecture                                             #
        #                                                                          #
        # HINT: nn.Sequential might be helpful.                                    #
        ############################################################################
        # *****START OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****

        self.model = nn.Sequential(
            nn.Conv2d(1,32,kernel_size=5,stride=1),
            nn.LeakyReLU(0.01),
            nn.MaxPool2d(kernel_size = 2, stride = 2),
            nn.Conv2d(32,64,kernel_size=5,stride=1),
            nn.LeakyReLU(0.01),
            nn.MaxPool2d(kernel_size = 2, stride = 2),
            nn.Flatten(),
            nn.Linear(4*4*64,4*4*64),
            nn.LeakyReLU(0.01),
            nn.Linear(4*4*64,1),
        )

        # *****END OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
        ############################################################################
        #                                END OF YOUR CODE                          #
        ############################################################################

    def forward(self, x):
        return self.model(x)


class DCGenerator(nn.Module):
    def __init__(self, noise_dim: int = NOISE_DIM):
        super().__init__()
        self.noise_dim = noise_dim

        ############################################################################
        # TODO: Implement architecture                                             #
        #                                                                          #
        # HINT: nn.Sequential might be helpful.                                    #
        ############################################################################
        # *****START OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****

        self.model = nn.Sequential(
```

```
            nn.Linear(self.noise_dim,1024),
            nn.ReLU(),
            nn.BatchNorm1d(1024),
            nn.Linear(1024,7*7*128),
            nn.ReLU(),
            nn.BatchNorm1d(7*7*128),
            nn.Unflatten(1, (128, 7, 7)),
            nn.ConvTranspose2d(128,64, kernel_size=4, stride = 2, padding=1),
            nn.ReLU(),
            nn.BatchNorm2d(64),
            nn.ConvTranspose2d(64,1, kernel_size=4, stride = 2, padding=1),
            nn.Tanh(),
            nn.Flatten()
        )

        # *****END OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
        ##############################################################################
        #                                END OF YOUR CODE                            #
        ##############################################################################

    def forward(self, x):
        return self.model(x)


def run_a_gan(
    D: nn.Module,
    G: nn.Module,
    D_solver: torch.optim.Optimizer,
    G_solver: torch.optim.Optimizer,
    discriminator_loss,
    generator_loss,
    loader_train: DataLoader,
    show_every: int = 250,
    batch_size: int = 128,
    noise_size: int = 96,
    num_epochs: int = 10,
) -> list[np.ndarray]:
    """
    Train a GAN!

    Inputs:
    - D, G: PyTorch models for the discriminator and generator
    - D_solver, G_solver: torch.optim Optimizers to use for training the
      discriminator and generator.
    - discriminator_loss, generator_loss: Functions to use for computing the generator and
      discriminator loss, respectively.
    - show_every: Show samples after every show_every iterations.
    - batch_size: Batch size to use for training.
    - noise_size: Dimension of the noise to use as input to the generator.
    - num_epochs: Number of epochs over the training dataset to use for training.
```

```
    """
    images = []
    iter_count = 0
    for epoch in range(num_epochs):
        for x, _ in loader_train:
            D_solver.zero_grad()
            real_data = x.to(device)
            logits_real = D(2 * (real_data - 0.5)).to(device)

            g_fake_seed = sample_noise(batch_size, noise_size).to(device)
            fake_images = G(g_fake_seed).detach()
            logits_fake = D(fake_images.view(batch_size, 1, 28, 28))

            d_total_error = discriminator_loss(logits_real, logits_fake)
            d_total_error.backward()
            D_solver.step()

            G_solver.zero_grad()
            g_fake_seed = sample_noise(batch_size, noise_size).to(device)
            fake_images = G(g_fake_seed)

            gen_logits_fake = D(fake_images.view(batch_size, 1, 28, 28))
            g_error = generator_loss(gen_logits_fake)
            g_error.backward()
            G_solver.step()

            if iter_count % show_every == 0:
                print(f'Iter: {iter_count}, D: {d_total_error.item():.4f}, G:
{g_error.item():.4f}')
                imgs_numpy = fake_images.data.cpu().numpy()
                images.append(imgs_numpy[0:16])

            iter_count += 1

    return images


def initialize_weights(m: nn.Module) -> None:
    if isinstance(m, nn.Linear) or isinstance(m, nn.ConvTranspose2d):
        nn.init.xavier_uniform_(m.weight.data)


def preprocess_img(x: Tensor) -> Tensor:
    return 2 * x - 1.0


def deprocess_img(x: Tensor) -> Tensor:
    return (x + 1.0) / 2.0


```

```
def rel_error(x: np.ndarray, y: np.ndarray) -> float:
    return np.max(np.abs(x - y) / (np.maximum(1e-8, np.abs(x) + np.abs(y))))


def count_params(model: nn.Module) -> int:
    """Count the number of parameters in the model."""
    return sum(p.numel() for p in model.parameters())

```

We would like to acknowledge Stanford University's CS231n on which we based the development of this project.

In [1]:
```
import os
import sys
sys.path.insert(0, os.path.dirname(os.path.abspath('gan.py')))
print('cwd:', os.getcwd())
```

```
cwd: /content
```

Enabling GPU

To enable GPU on your gcloud instance, make sure you selected a GPU machine type when creating your VM. You can verify the GPU is available by running `nvidia-smi` in the terminal.

Generative Adversarial Networks (GANs)

In C147/C247, all the applications of neural networks that we have explored have been **discriminative models** that take an input and are trained to produce a labeled output. In this notebook, we will expand our repetoire, and build **generative models** using neural networks. Specifically, we will learn how to build models which generate novel images that resemble a set of training images.

What is a GAN?

In 2014, Goodfellow et al. presented a method for training generative models called Generative Adversarial Networks (GANs for short). In a GAN, we build two different neural networks. Our first network is a traditional classification network, called the **discriminator**. We will train the discriminator to take images and classify them as being real (belonging to the training set) or fake (not present in the training set). Our other network, called the **generator**, will take random noise as input and transform it using a neural network to produce images. The goal of the generator is to fool the discriminator into thinking the images it produced are real.

We can think of this back and forth process of the generator (G) trying to fool the discriminator (D) and the discriminator trying to correctly classify real vs. fake as a minimax game:

$$\underset{G}{\text{minimize}}\ \underset{D}{\text{maximize}}\ \mathbb{E}_{x\sim p_{\text{data}}}\left[\log D(x)\right] + \mathbb{E}_{z\sim p(z)}\left[\log(1-D(G(z)))\right]$$

where $z \sim p(z)$ are the random noise samples, $G(z)$ are the generated images using the neural network generator G, and D is the output of the discriminator, specifying the probability of an input being real. In Goodfellow et al., they analyze this minimax game and show how it relates to minimizing the Jensen-Shannon divergence between the training data distribution and the generated samples from G.

To optimize this minimax game, we will alternate between taking gradient *descent* steps on the objective for G and gradient *ascent* steps on the objective for D:

1. update the **generator** (G) to minimize the probability of the **discriminator making the correct choice**.
2. update the **discriminator** (D) to maximize the probability of the **discriminator making the correct choice**.

While these updates are useful for analysis, they do not perform well in practice. Instead, we will use a different objective when we update the generator: maximize the probability of the **discriminator making the incorrect choice**. This small change helps to allevaiate problems with the generator gradient vanishing when the discriminator is confident. This is the standard update used in most GAN papers and was used in the original paper from Goodfellow et al..

In this assignment, we will alternate the following updates:

1. Update the generator (G) to maximize the probability of the discriminator making the incorrect choice on generated data:

$$\underset{G}{\text{maximize}}\ \mathbb{E}_{z\sim p(z)}\left[\log D(G(z))\right]$$

2. Update the discriminator (D), to maximize the probability of the discriminator making the correct choice on real and generated data:

$$\underset{D}{\text{maximize}}\ \mathbb{E}_{x\sim p_{\text{data}}}\left[\log D(x)\right] + \mathbb{E}_{z\sim p(z)}\left[\log(1-D(G(z)))\right]$$

Here's an example of what your outputs from the 3 different models you're going to train should look like. Note that GANs are sometimes finicky, so your outputs might not look exactly like this. This is just meant to be a *rough* guideline of the kind of quality you can expect:

Assignment Overview

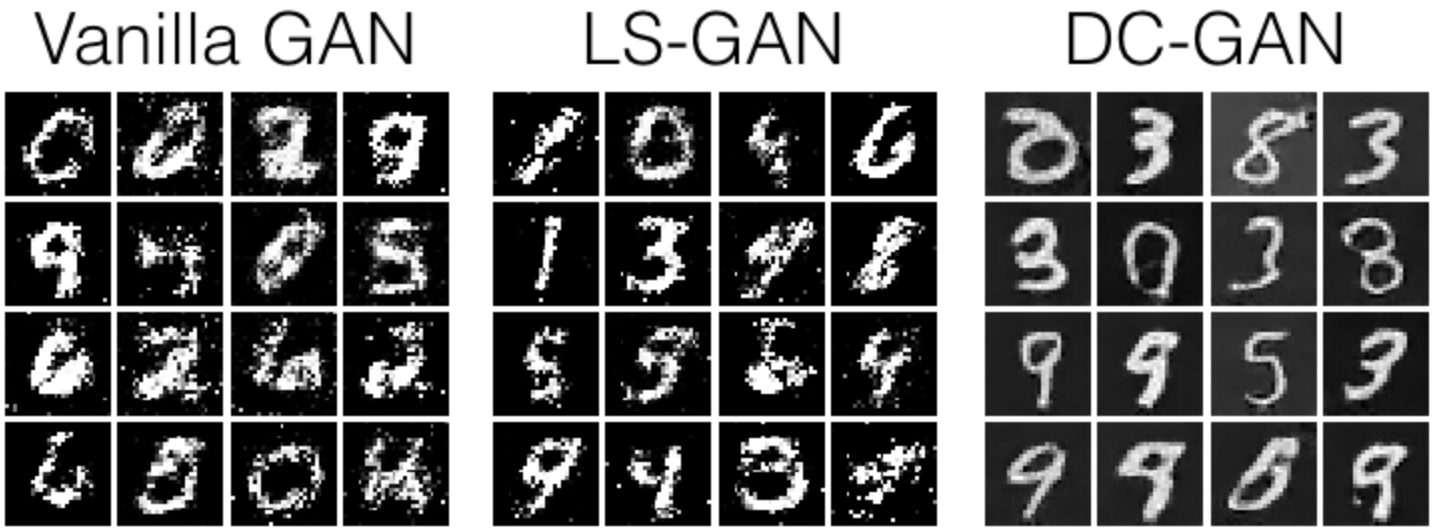
All your code goes in `gan.py`. The notebook imports from it and runs checks — do not modify the notebook cells.

Task	Points
`Discriminator` — MLP architecture	1
`Generator` — MLP architecture	1
`discriminator_loss` + `generator_loss` — BCE GAN loss	2
`ls_discriminator_loss` + `ls_generator_loss` — LSGAN loss	2
`DCDiscriminator` — convolutional architecture	1
`DCGenerator` — convolutional architecture	1
Inline Questions 4, 5, 6	3
Total	**11**

`sample_noise`, `get_optimizer`, `bce_loss`, and `run_a_gan` are already implemented for you.

In [2]:
```
# Run this cell to see sample outputs.
from IPython.display import Image
Image('assets/gan_outputs.png')
```

Out[2]:



In [3]:

```
# Setup cell.
import numpy as np
import torch
import torch.nn as nn
import torchvision.transforms as T
import torch.optim as optim
from torch.utils.data import DataLoader
import torchvision.datasets as dset
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
from gan import rel_error, count_params

%matplotlib inline
plt.rcParams['figure.figsize'] = (10.0, 8.0)
plt.rcParams['image.interpolation'] = 'nearest'
plt.rcParams['image.cmap'] = 'gray'

import importlib
import sys
sys.modules["imp"] = importlib

%load_ext autoreload
%autoreload 2

def show_images(images):
    images = np.reshape(images, [images.shape[0], -1])
    sqrtn = int(np.ceil(np.sqrt(images.shape[0])))
    sqrtimg = int(np.ceil(np.sqrt(images.shape[1])))

    fig = plt.figure(figsize=(sqrtn, sqrtn))
    gs = gridspec.GridSpec(sqrtn, sqrtn)
    gs.update(wspace=0.05, hspace=0.05)

    for i, img in enumerate(images):
        ax = plt.subplot(gs[i])
        plt.axis('off')
        ax.set_xticklabels([])
        ax.set_yticklabels([])
        ax.set_aspect('equal')
        plt.imshow(img.reshape([sqrtimg, sqrtimg]))
    return
```

```
answers = dict(np.load('assets/gan-checks.npz'))
device = torch.device('cuda') if torch.cuda.is_available() else (torch.device('mps') if torch.backends.mps.is_availab
print(device)
```

```
cuda
```

Dataset

GANs are notoriously finicky with hyperparameters, and also require many training epochs. In order to make this assignment approachable, we will be working on the MNIST dataset, which is 60,000 training and 10,000 test images. Each picture contains a centered image of white digit on black background (0 through 9). This was one of the first datasets used to train convolutional neural networks and it is fairly easy -- a standard CNN model can easily exceed 99% accuracy.

To simplify our code here, we will use the PyTorch MNIST wrapper, which downloads and loads the MNIST dataset. See the documentation for more information about the interface. The data will be saved into a folder called `MNIST`.

In [4]:
```
NOISE_DIM = 96
batch_size = 128

mnist_train = dset.MNIST('./mnist_data', train=True, download=True, transform=T.ToTensor())
loader_train = DataLoader(mnist_train, batch_size=batch_size, shuffle=True, drop_last=True)

iterator = iter(loader_train)
imgs, labels = next(iterator)
imgs = imgs.view(batch_size, 784).numpy().squeeze()
show_images(imgs)
```

```
100%|██████████| 9.91M/9.91M [00:00<00:00, 40.8MB/s]
100%|██████████| 28.9k/28.9k [00:00<00:00, 1.13MB/s]
100%|██████████| 1.65M/1.65M [00:00<00:00, 10.4MB/s]
100%|██████████| 4.54k/4.54k [00:00<00:00, 26.8MB/s]
```

PyTorch Utilities

For the convolutional generator you will need to reshape tensors between flat vectors and image tensors. Use PyTorch's built-in `nn.Flatten()` and `nn.Unflatten(dim, shape)` — no need to import anything extra.

We also provide a weight initializer (and call it for you) that uses Xavier initialization instead of PyTorch's uniform default.

In [5]:
```
from gan import initialize_weights
```

Discriminator (1 point)

Our first step is to build a discriminator. Fill in `self.model` as an `nn.Sequential` inside the `Discriminator` class in `gan.py`. All fully connected layers should include bias terms. The architecture is:

- Fully connected layer with input size 784 and output size 256
- LeakyReLU with alpha 0.01
- Fully connected layer with input_size 256 and output size 256
- LeakyReLU with alpha 0.01
- Fully connected layer with input size 256 and output size 1

Recall that the Leaky ReLU nonlinearity computes $f(x) = \max(\alpha x, x)$ for some fixed constant α; for the LeakyReLU nonlinearities in the architecture above we set $\alpha = 0.01$.

The output of the discriminator should have shape `[batch_size, 1]`, and contain real numbers corresponding to the scores that each of the `batch_size` inputs is a real image.

Implement `Discriminator` in `gan.py`

Test to make sure the number of parameters in the discriminator is correct:

In [6]:
```
from gan import Discriminator

def test_discriminator(true_count=267009):
    model = Discriminator()
    cur_count = count_params(model)
    if cur_count != true_count:
        print('Incorrect number of parameters in discriminator. Check your architecture.')
    else:
        print('Correct number of parameters in discriminator.')

test_discriminator()
```

```
Correct number of parameters in discriminator.
```

Generator (1 point)

Now to build the generator network. Fill in `self.model` as an `nn.Sequential` inside the `Generator` class in `gan.py`:

- Fully connected layer from noise_dim to 1024
- `ReLU`
- Fully connected layer with size 1024
- `ReLU`
- Fully connected layer with size 784
- `TanH` (to clip the image to be in the range of [-1,1])

All fully connected layers should include bias terms. Implement `Generator` in `gan.py`

Test to make sure the number of parameters in the generator is correct:

In [7]:
```
from gan import Generator

def test_generator(true_count=1858320):
    model = Generator(noise_dim=4)
    cur_count = count_params(model)
    if cur_count != true_count:
        print('Incorrect number of parameters in generator. Check your architecture.')
    else:
```

```
        print('Correct number of parameters in generator.')

test_generator()
```

```
Correct number of parameters in generator.
```

GAN Loss (2 points)

Compute the generator and discriminator loss. The generator loss is:

$$\ell_G = -\mathbb{E}_{z\sim p(z)}\left[\log D(G(z))\right]$$

and the discriminator loss is:

$$\ell_D = -\mathbb{E}_{x\sim p_{\text{data}}}\left[\log D(x)\right] - \mathbb{E}_{z\sim p(z)}\left[\log(1 - D(G(z)))\right]$$

Note that these are negated from the equations presented earlier as we will be *minimizing* these losses.

HINTS: You should use the `bce_loss` function defined below to compute the binary cross entropy loss which is needed to compute the log probability of the true label given the logits output from the discriminator. Given a score $s \in \mathbb{R}$ and a label $y \in \{0, 1\}$, the binary cross entropy loss is

$$bce(s, y) = -y * \log(s) - (1 - y) * \log(1 - s)$$

A naive implementation of this formula can be numerically unstable, so we have provided a numerically stable implementation that relies on PyTorch's `nn.BCEWithLogitsLoss`.

You will also need to compute labels corresponding to real or fake and use the logit arguments to determine their size. Make sure you use the `torch.float32` data type (same as `torch.float`) for compatibility with BCEWithLogitsLoss, which uses floating point data types, and move tensors to GPU (if applicable) using `u = u.to(device)`.

`true_labels = torch.ones(size, dtype=torch.float32).to(device)`

or

`true_labels = torch.ones(size, device=device) # uses torch.float32 by default`

Instead of computing the expectation of $\log D(G(z))$, $\log D(x)$ and $\log(1 - D(G(z)))$, we will be averaging over elements of the minibatch. This is taken care of in `bce_loss` which combines the loss by averaging. **Hint**: Do NOT concatenate real and fake losses. Sum the two (empirical) expectations instead.

Implement `discriminator_loss` and `generator_loss` in `gan.py`

Test your generator and discriminator loss. You should see errors < 1e-7.

In [8]:
```
from gan import bce_loss, discriminator_loss, generator_loss

def test_discriminator_loss(logits_real, logits_fake, d_loss_true):
    d_loss = discriminator_loss(
        torch.tensor(logits_real, dtype=torch.float32).to(device),
        torch.tensor(logits_fake, dtype=torch.float32).to(device)
    ).cpu().numpy()
    print(f'Maximum error in d_loss: {rel_error(d_loss_true, d_loss):g}')

test_discriminator_loss(
    answers['logits_real'],
    answers['logits_fake'],
    answers['d_loss_true']
)
```

```
Maximum error in d_loss: 3.97058e-09
```

In [9]:
```
def test_generator_loss(logits_fake, g_loss_true):
    g_loss = generator_loss(
        torch.tensor(logits_fake, dtype=torch.float32).to(device)
    ).cpu().numpy()
    print(f'Maximum error in g_loss: {rel_error(g_loss_true, g_loss):g}')

test_generator_loss(
    answers['logits_fake'],
    answers['g_loss_true']
)
```

```
Maximum error in g_loss: 3.4188e-08
```

Training a GAN!

We provide you the main training loop. You won't need to change `run_a_gan` in `gan.py`, but we encourage you to read through it for your own understanding. Training takes about 7 minutes on CPU and about 1-2 minutes on GPU.

If it doesn't work the first time, check the discriminator: Did you use `nn.Flatten()`?

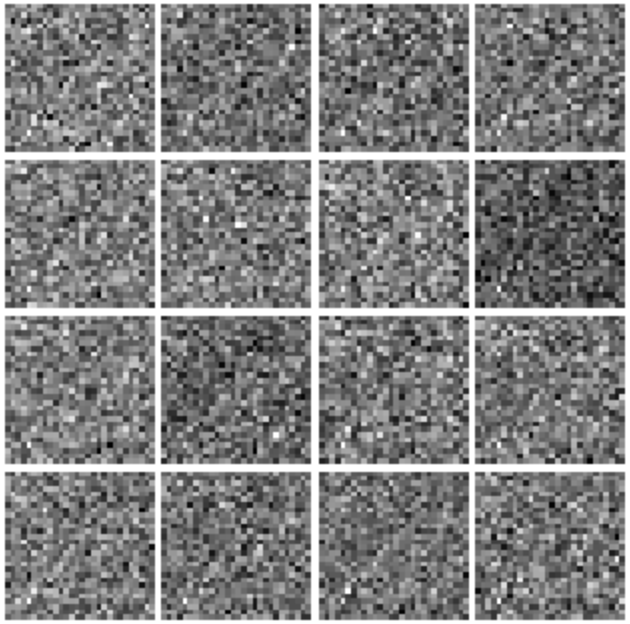
In [10]:
```
from gan import get_optimizer, run_a_gan

D = Discriminator().to(device)
G = Generator().to(device)

D_solver = get_optimizer(D)
G_solver = get_optimizer(G)

images = run_a_gan(
    D,
    G,
    D_solver,
    G_solver,
    discriminator_loss,
    generator_loss,
    loader_train
)
```

```
Iter: 0, D: 1.4403, G: 0.7045
Iter: 250, D: 1.6972, G: 0.8097
Iter: 500, D: 1.3028, G: 0.8934
Iter: 750, D: 1.4852, G: 0.8930
Iter: 1000, D: 1.3281, G: 1.1697
Iter: 1250, D: 1.0929, G: 1.1373
Iter: 1500, D: 1.1950, G: 0.9230
Iter: 1750, D: 1.1540, G: 1.0410
Iter: 2000, D: 1.3149, G: 1.0300
Iter: 2250, D: 1.2292, G: 0.7668
Iter: 2500, D: 1.3245, G: 0.8675
Iter: 2750, D: 1.3109, G: 0.8069
Iter: 3000, D: 1.3146, G: 0.8347
Iter: 3250, D: 1.3411, G: 0.8518
Iter: 3500, D: 1.3710, G: 0.7313
Iter: 3750, D: 1.3100, G: 0.8249
Iter: 4000, D: 1.3197, G: 0.7320
Iter: 4250, D: 1.3356, G: 0.7085
Iter: 4500, D: 1.2977, G: 0.8053
```

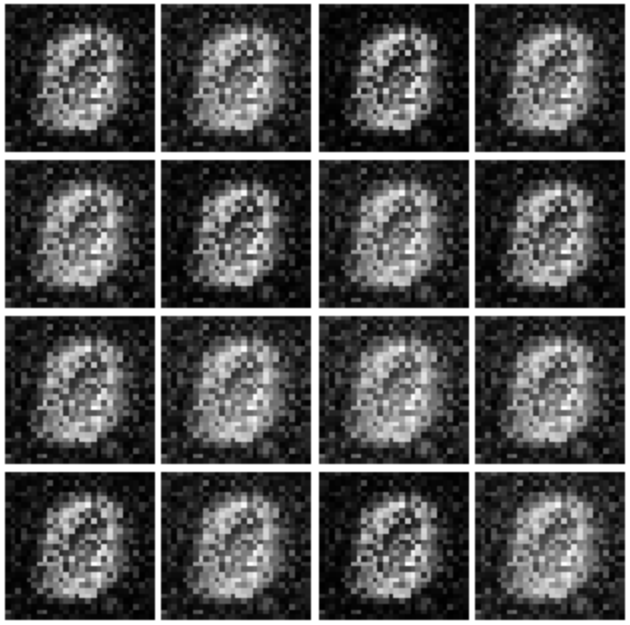
Run the cell below to show the generated images.

In [11]:
```python
numIter = 0
for img in images:
    print(f'Iter: {numIter}')
    show_images(img)
    plt.show()
    numIter += 250
    print()
```

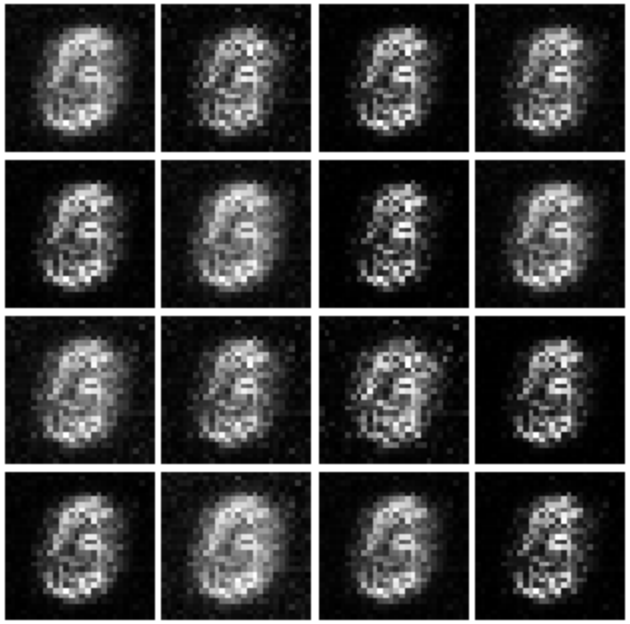
```
Iter: 0
```



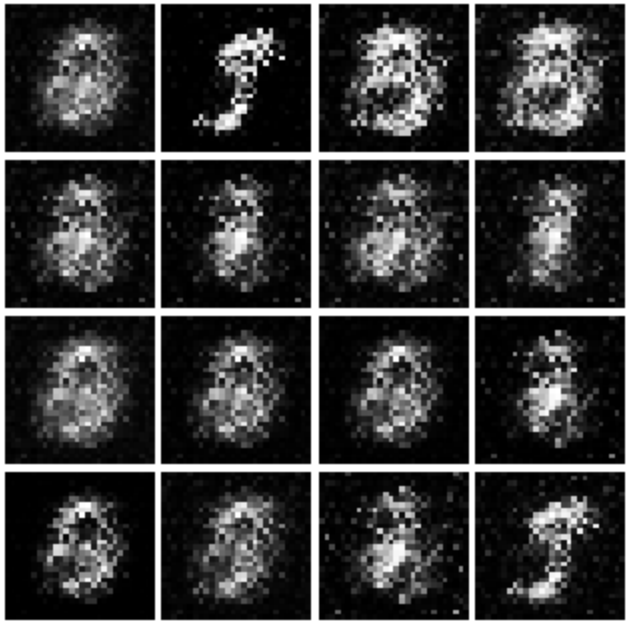
Iter: 250



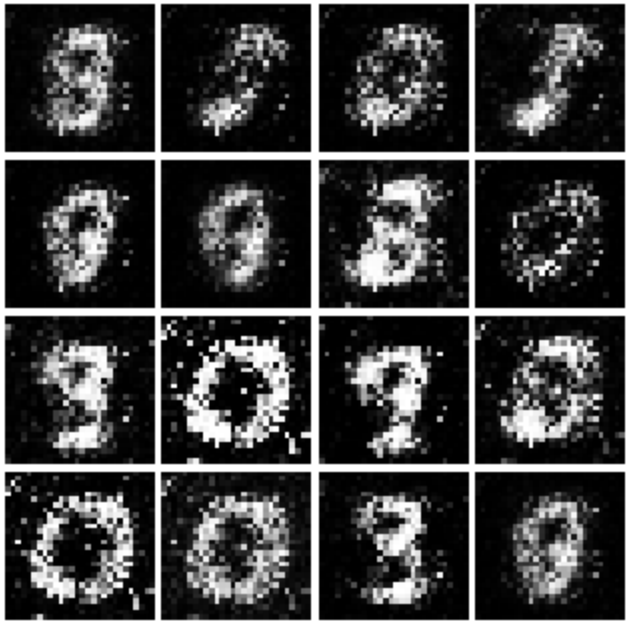
Iter: 500



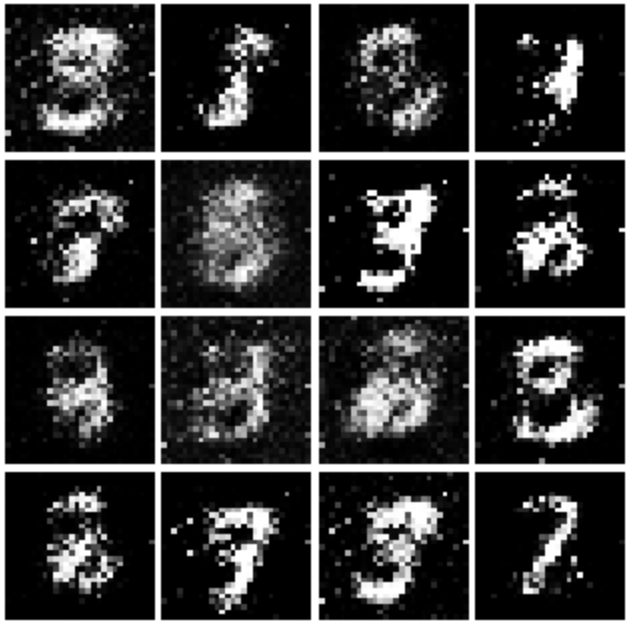
Iter: 750



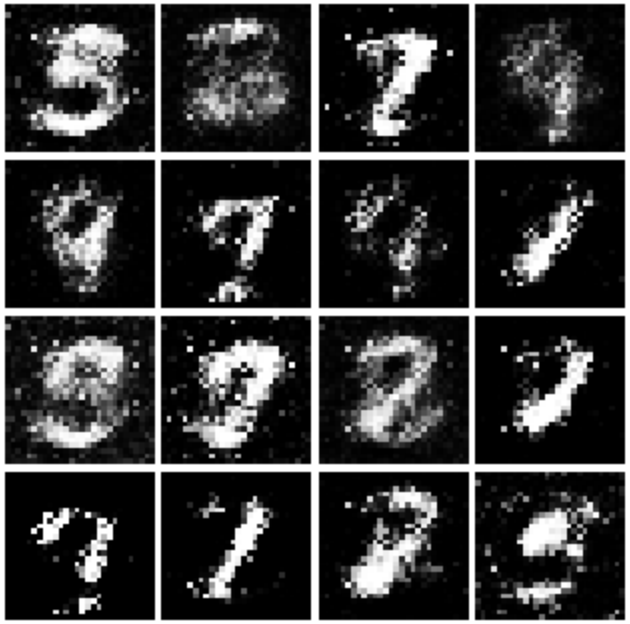
Iter: 1000



Iter: 1250

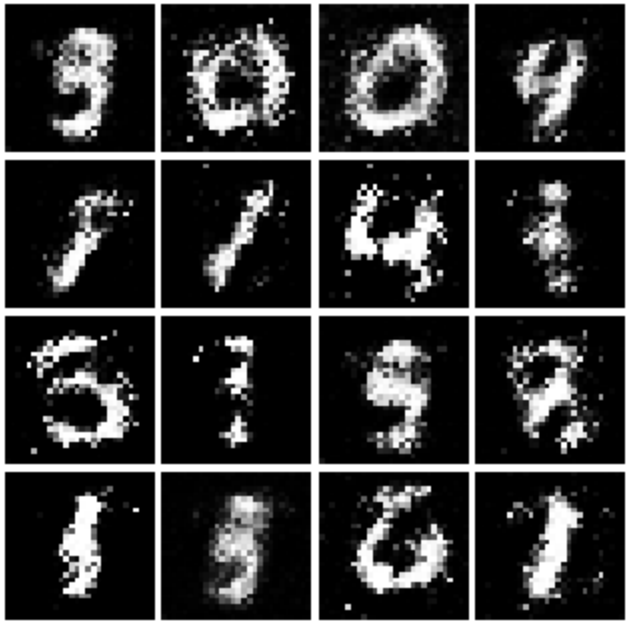


Iter: 1500



Iter: 1750

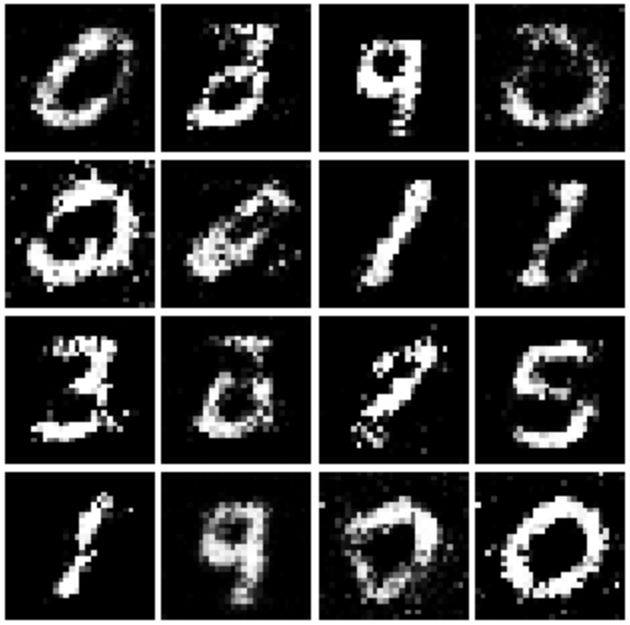
Iter: 2000



Iter: 2250

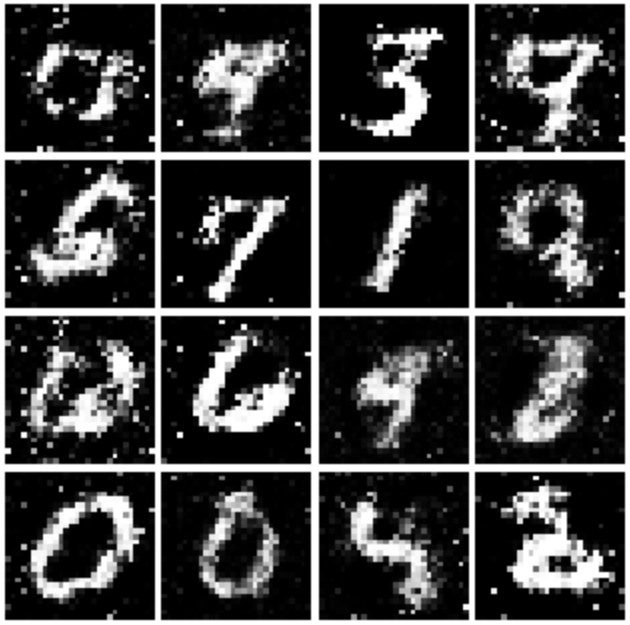
Iter: 2500

Iter: 2750



Iter: 3000

Iter: 3250

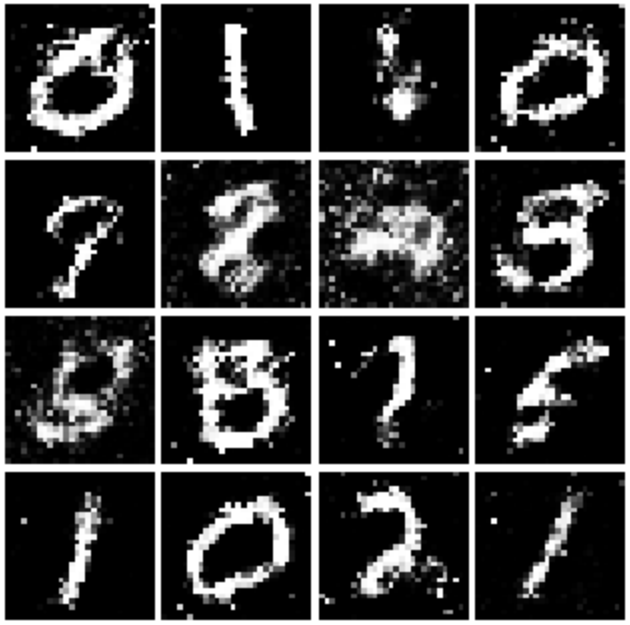


Iter: 3500

Iter: 3750

Iter: 4000

Iter: 4250



Iter: 4500

Inline Question 1

What does your final vanilla GAN image look like?

In [12]:
```
# This output is your answer.
print('Vanilla GAN final image:')
show_images(images[-1])
plt.show()
```

```
Vanilla GAN final image:
```

Well that wasn't so hard, was it? In the iterations in the low 100s you should see black backgrounds, fuzzy shapes as you approach iteration 1000, and decent shapes, about half of which will be sharp and clearly recognizable as we pass 3000.

Least Squares GAN (2 points)

We'll now look at Least Squares GAN, a newer, more stable alernative to the original GAN loss function. For this part, all we have to do is change the loss function and retrain the model. We'll implement equation (9) in the paper, with the generator loss:

$$\ell_G = \frac{1}{2}\mathbb{E}_{z\sim p(z)}\left[\left(D(G(z))-1\right)^2\right]$$

and the discriminator loss:

$$\ell_D = \frac{1}{2}\mathbb{E}_{x\sim p_{\text{data}}}\left[\left(D(x)-1\right)^2\right] + \frac{1}{2}\mathbb{E}_{z\sim p(z)}\left[\left(D(G(z))\right)^2\right]$$

HINTS: Instead of computing the expectation, we will be averaging over elements of the minibatch, so make sure to combine the loss by averaging instead of summing. When plugging in for $D(x)$ and $D(G(z))$ use the direct output from the discriminator

(`scores_real` and `scores_fake`). **If** you get errors of 0.333..., you probably forgot to include the 1/2 term.

Implement `ls_discriminator_loss` , `ls_generator_loss` in `gan.py`

Before running a GAN with our new loss function, let's check it:

In [13]:
```
from gan import ls_discriminator_loss, ls_generator_loss

def test_lsgan_loss(score_real, score_fake, d_loss_true, g_loss_true):
    score_real = torch.tensor(score_real, dtype=torch.float32).to(device)
    score_fake = torch.tensor(score_fake, dtype=torch.float32).to(device)
    d_loss = ls_discriminator_loss(score_real, score_fake).cpu().numpy()
    g_loss = ls_generator_loss(score_fake).cpu().numpy()
    print(f'Maximum error in d_loss: {rel_error(d_loss_true, d_loss):g}')
    print(f'Maximum error in g_loss: {rel_error(g_loss_true, g_loss):g}')

test_lsgan_loss(
    answers['logits_real'],
    answers['logits_fake'],
    answers['d_loss_lsgan_true'],
    answers['g_loss_lsgan_true']
)
```

```
Maximum error in d_loss: 1.53171e-08
Maximum error in g_loss: 2.7837e-09
```

Run the following cell to train your model! Training takes about 7 minutes on CPU and about 1-2 minutes on GPU.

In [14]:
```
D_LS = Discriminator().to(device)
G_LS = Generator().to(device)

D_LS_solver = get_optimizer(D_LS)
G_LS_solver = get_optimizer(G_LS)

images = run_a_gan(
    D_LS,
    G_LS,
    D_LS_solver,
    G_LS_solver,
    ls_discriminator_loss,
    ls_generator_loss,
```

```
    loader_train
)
```

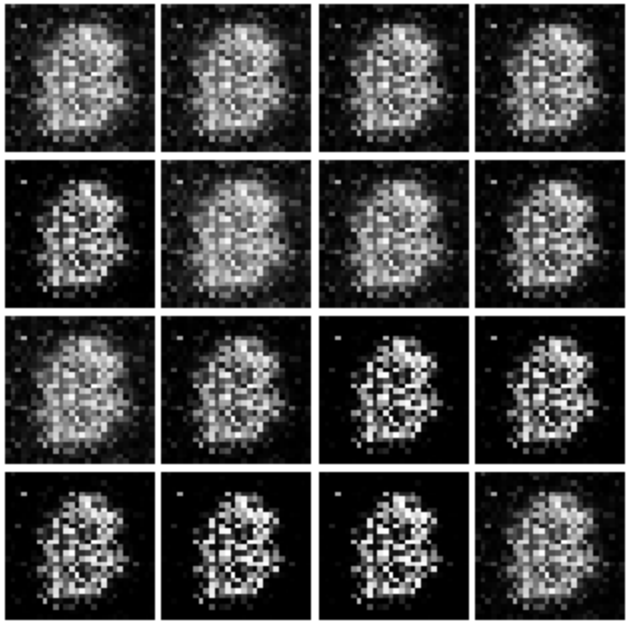
```
Iter: 0, D: 0.5976, G: 0.4566
Iter: 250, D: 0.1749, G: 0.4213
Iter: 500, D: 0.1942, G: 0.1779
Iter: 750, D: 0.1565, G: 0.3126
Iter: 1000, D: 0.1485, G: 0.2529
Iter: 1250, D: 0.2536, G: 0.2709
Iter: 1500, D: 0.2033, G: 0.1523
Iter: 1750, D: 0.2152, G: 0.2086
Iter: 2000, D: 0.2351, G: 0.1653
Iter: 2250, D: 0.2197, G: 0.1796
Iter: 2500, D: 0.2222, G: 0.1661
Iter: 2750, D: 0.2202, G: 0.1699
Iter: 3000, D: 0.2215, G: 0.1504
Iter: 3250, D: 0.2370, G: 0.1602
Iter: 3500, D: 0.2147, G: 0.1787
Iter: 3750, D: 0.2260, G: 0.1718
Iter: 4000, D: 0.2300, G: 0.1624
Iter: 4250, D: 0.2253, G: 0.1416
Iter: 4500, D: 0.2285, G: 0.1455
```

Run the cell below to show generated images.

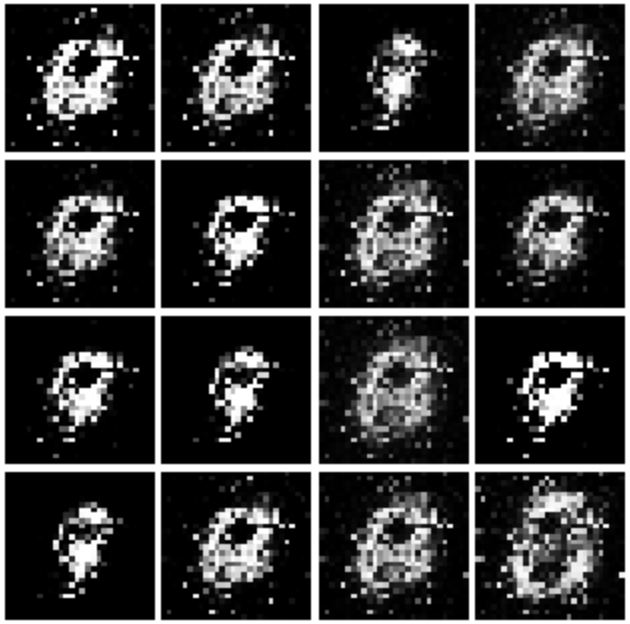
In [15]:
```
numIter = 0
for img in images:
    print(f'Iter: {numIter}')
    show_images(img)
    plt.show()
    numIter += 250
    print()
```

```
Iter: 0
```

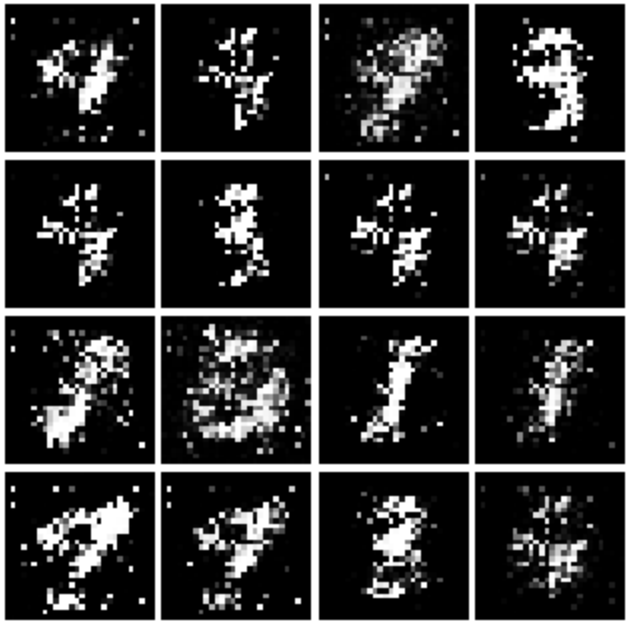
Iter: 250



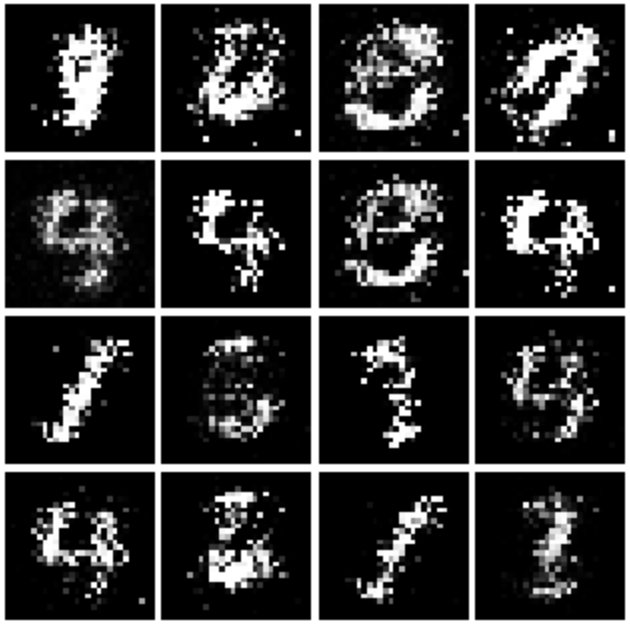
Iter: 500



Iter: 750

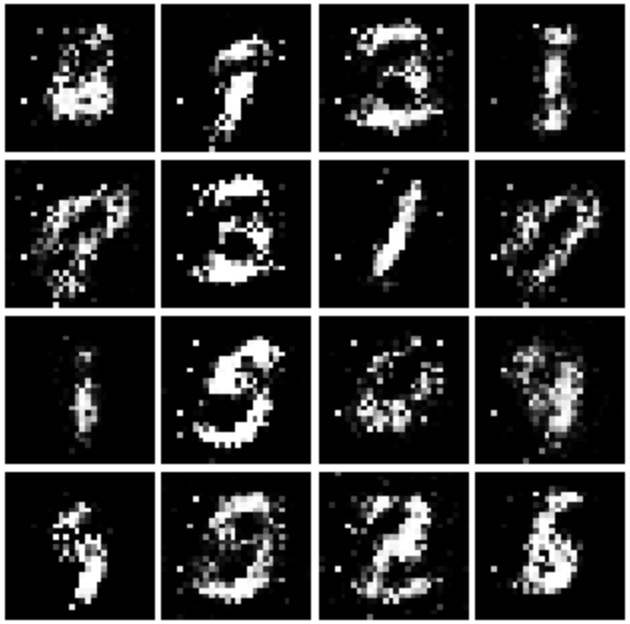


Iter: 1000

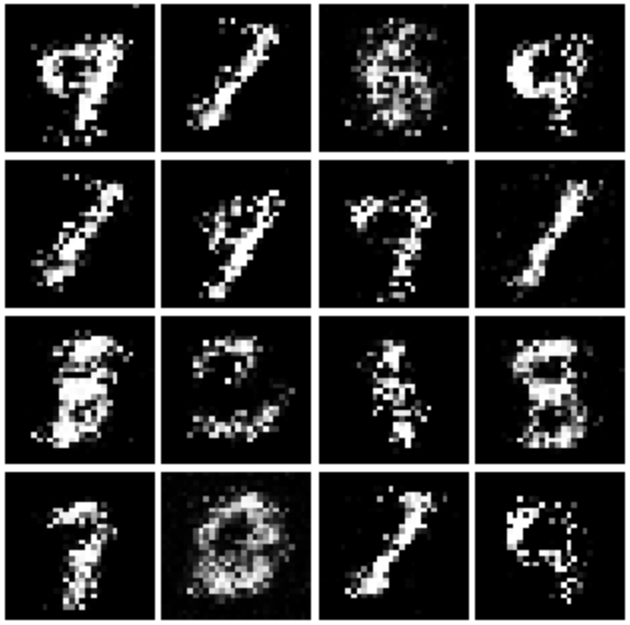


Iter: 1250

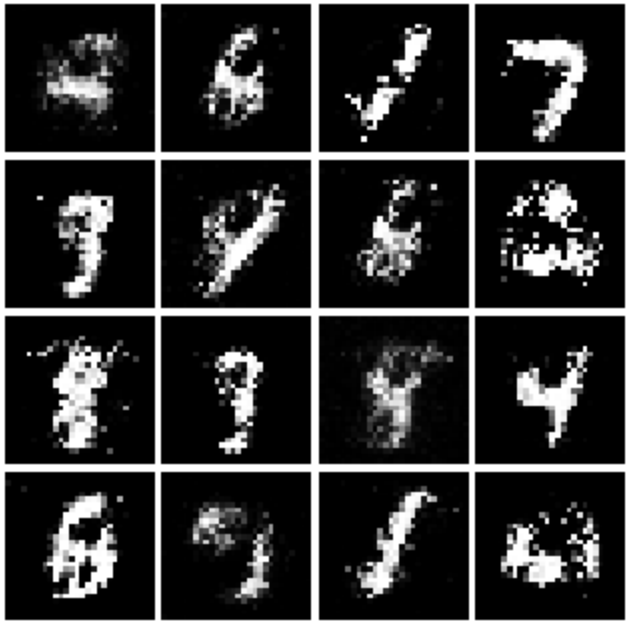
Iter: 1500



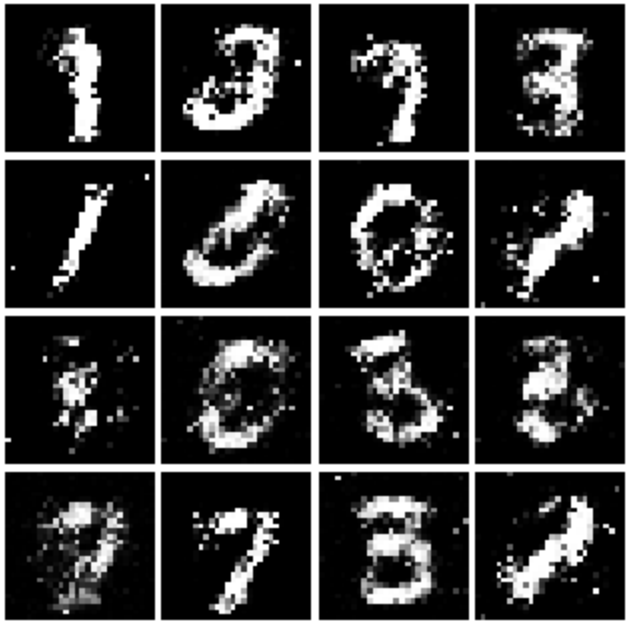
Iter: 1750



Iter: 2000



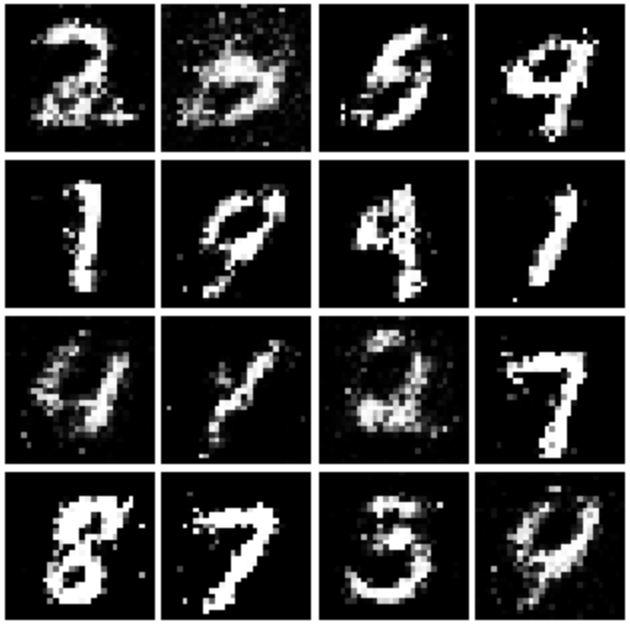
Iter: 2250



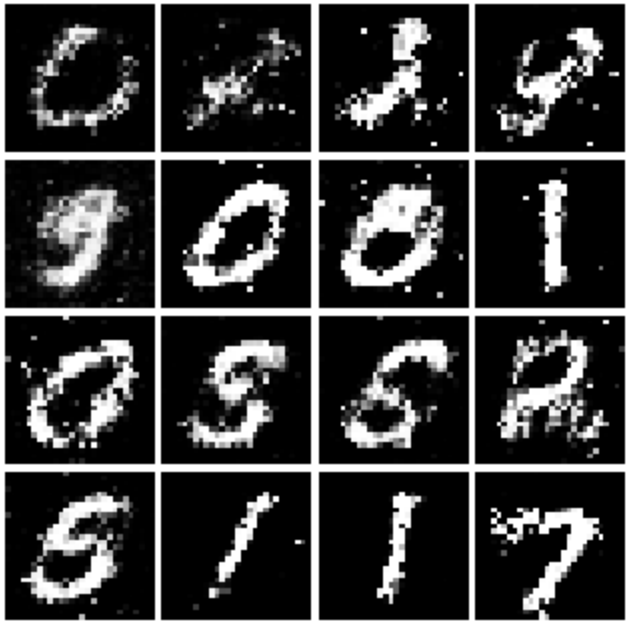
Iter: 2500

Iter: 2750

Iter: 3000



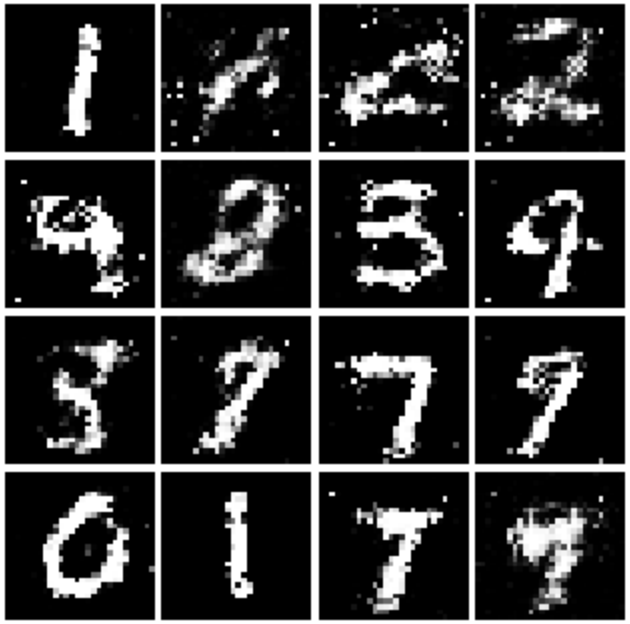
Iter: 3250



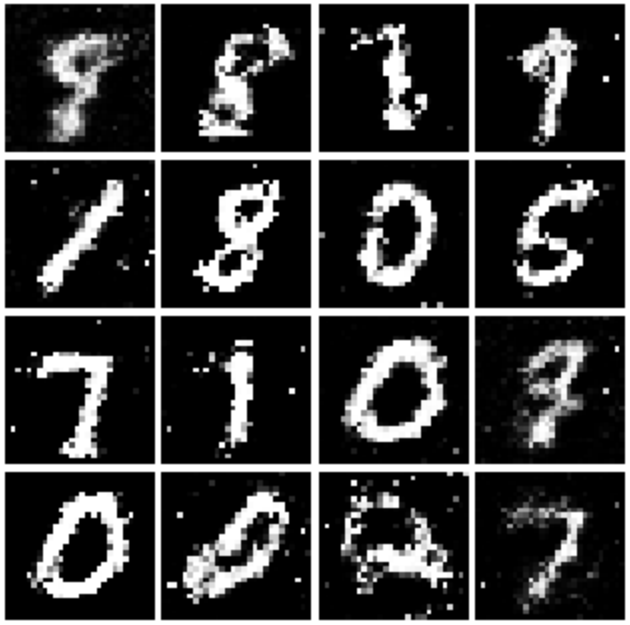
Iter: 3500

Iter: 3750

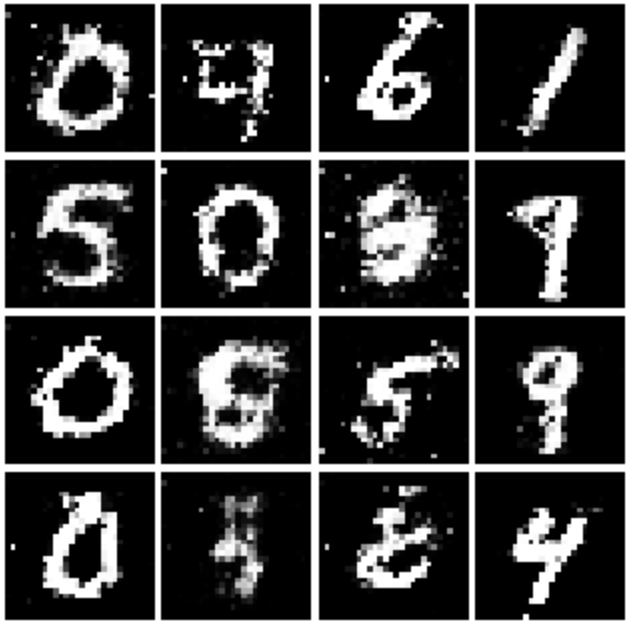
Iter: 4000



Iter: 4250



Iter: 4500

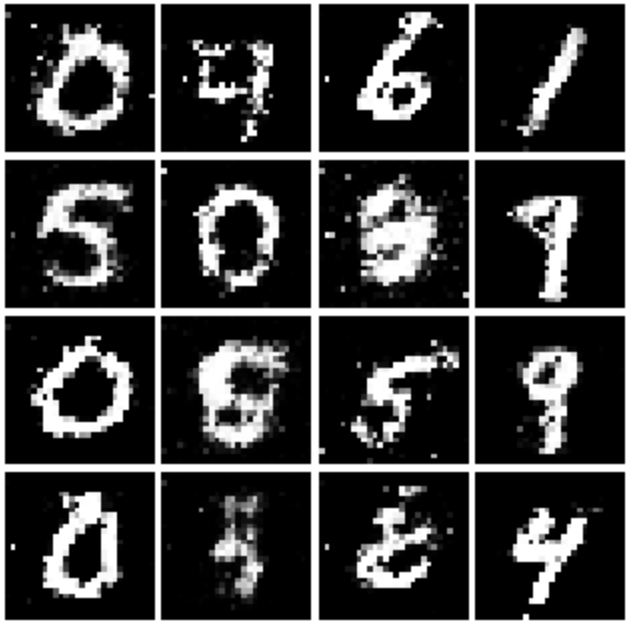


Inline Question 2

What does your final LSGAN image look like?

In [16]:
```
# This output is your answer.
print('LSGAN final image:')
show_images(images[-1])
plt.show()
```

```
LSGAN final image:
```



Deep Convolutional GAN (2 points)

In the first part of the notebook, we implemented an almost direct copy of the original GAN network from Ian Goodfellow. However, this network architecture allows no real spatial reasoning. It is unable to reason about things like "sharp edges" in general because it lacks any convolutional layers. Thus, in this section, we will implement some of the ideas from DCGAN, where we use convolutional networks

Discriminator

We will use a discriminator inspired by the TensorFlow MNIST classification tutorial, which is able to get above 99% accuracy on the MNIST dataset fairly quickly. **Hint**: There is no need to specify padding. **Warning**: If you use LazyLinear I will take off points.

- Conv2D: 32 Filters, 5x5, Stride 1
- Leaky ReLU(alpha=0.01)
- Max Pool 2x2, Stride 2
- Conv2D: 64 Filters, 5x5, Stride 1

- Leaky ReLU(alpha=0.01)
- Max Pool 2x2, Stride 2
- Flatten
- Fully Connected with output size 4 x 4 x 64
- Leaky ReLU(alpha=0.01)
- Fully Connected with output size 1

Implement `DCDiscriminator` in `gan.py`

In [17]:
```
from gan import DCDiscriminator

data = next(enumerate(loader_train))[-1][0].to(device)
b = DCDiscriminator().to(device)
out = b(data)
print(out.size())
```

```
torch.Size([128, 1])
```

Check the number of parameters in your classifier as a sanity check:

In [18]:
```
def test_dc_classifer(true_count=1102721):
    model = DCDiscriminator()
    cur_count = count_params(model)
    if cur_count != true_count:
        print('Incorrect number of parameters in classifier. Check your architecture.')
    else:
        print('Correct number of parameters in classifier.')

test_dc_classifer()
```

```
Correct number of parameters in classifier.
```

Generator

For the generator, we will copy the architecture exactly from the InfoGAN paper. See Appendix C.1 MNIST. See the documentation for nn.ConvTranspose2d. We are always "training" in GAN mode.

- Fully connected with output size 1024
- `ReLU`

- BatchNorm
- Fully connected with output size 7 x 7 x 128
- ReLU
- BatchNorm
- Use `nn.Unflatten(1, (128, 7, 7))` to reshape into Image Tensor of shape 128, 7, 7
- `ConvTranspose2d`: 64 filters of 4x4, stride 2, 'same' padding (use `padding=1`)
- `ReLU`
- BatchNorm
- `ConvTranspose2d`: 1 filter of 4x4, stride 2, 'same' padding (use `padding=1`)
- `TanH`
- Should have a 28x28x1 image, reshape back into 784 vector (using `nn.Flatten()`)

Implement `DCGenerator` in `gan.py`

In [19]:
```
from gan import DCGenerator

test_g_gan = DCGenerator().to(device)
test_g_gan.apply(initialize_weights)

fake_seed = torch.randn(batch_size, NOISE_DIM).to(device)
fake_images = test_g_gan.forward(fake_seed)
fake_images.size()
```

Out[19]:
```
torch.Size([128, 784])
```

Check the number of parameters in your generator as a sanity check:

In [20]:
```
def test_dc_generator(true_count=6580801):
    model = DCGenerator(noise_dim=4)
    cur_count = count_params(model)
    if cur_count != true_count:
        print('Incorrect number of parameters in generator. Check your architecture.')
    else:
        print('Correct number of parameters in generator.')

test_dc_generator()
```

```
Correct number of parameters in generator.
```

Run the following cell to train your DCGAN. Training takes about 35 minutes on CPU and about 1 minute on GPU.

Note: DCGAN uses the original BCE loss (`discriminator_loss` / `generator_loss`), not LSGAN. The architectural improvements in DCGAN (convolutional layers, batch norm) are what stabilize training here — LSGAN is a separate axis of improvement that you could combine with DCGAN, but for this assignment we keep them separate.

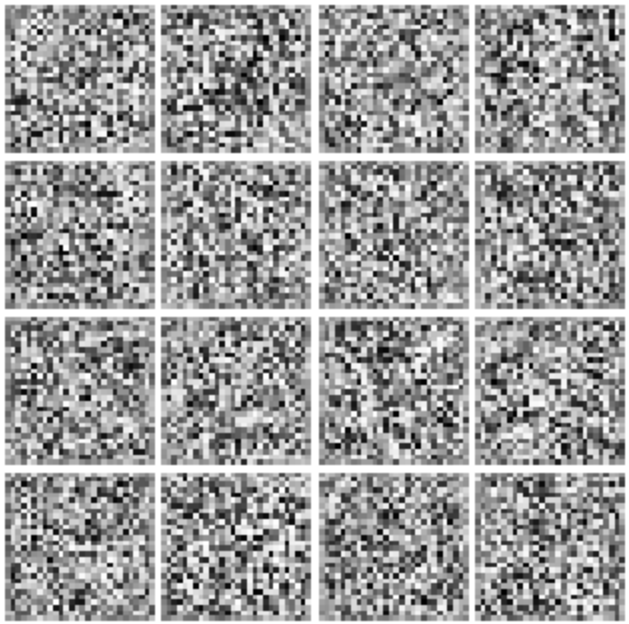
In [21]:
```
D_DC = DCDiscriminator().to(device)
D_DC.apply(initialize_weights)
G_DC = DCGenerator().to(device)
G_DC.apply(initialize_weights)

D_DC_solver = get_optimizer(D_DC)
G_DC_solver = get_optimizer(G_DC)

images = run_a_gan(
    D_DC,
    G_DC,
    D_DC_solver,
    G_DC_solver,
    discriminator_loss,
    generator_loss,
    loader_train,
    num_epochs=5
)
```

```
Iter: 0, D: 1.3322, G: 1.4283
Iter: 250, D: 1.0421, G: 0.9529
Iter: 500, D: 1.0899, G: 0.9975
Iter: 750, D: 1.0230, G: 1.2119
Iter: 1000, D: 1.0860, G: 1.3163
Iter: 1250, D: 1.1652, G: 0.9233
Iter: 1500, D: 1.2019, G: 1.6452
Iter: 1750, D: 1.0532, G: 1.2875
Iter: 2000, D: 1.0446, G: 0.8433
Iter: 2250, D: 1.2349, G: 0.6944
```

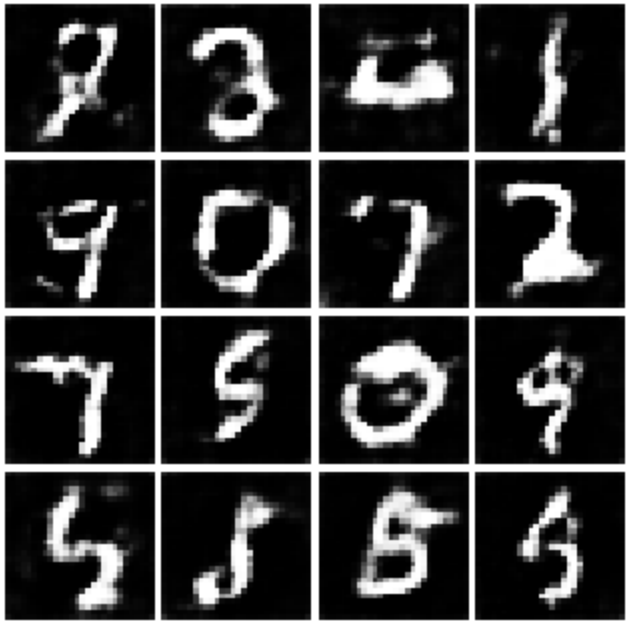
Run the cell below to show generated images.

In [22]:
```
numIter = 0
for img in images:
    print(f'Iter: {numIter}')
    show_images(img)
    plt.show()
    numIter += 250
    print()
```

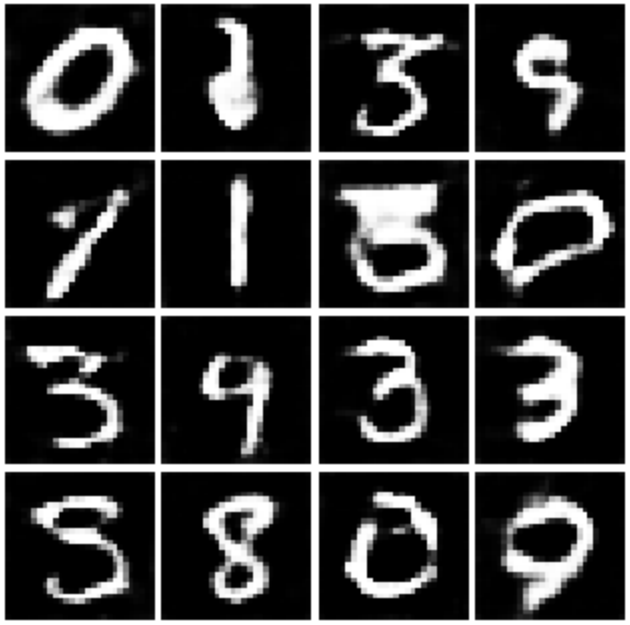
```
Iter: 0
```



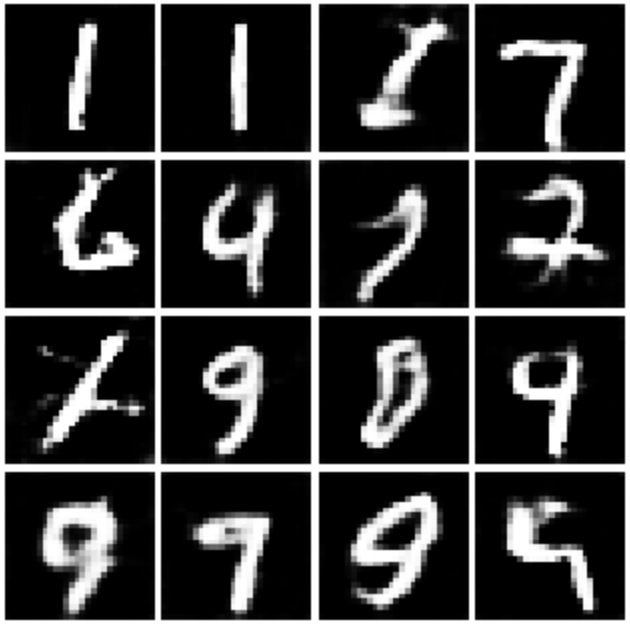
```
Iter: 250
```



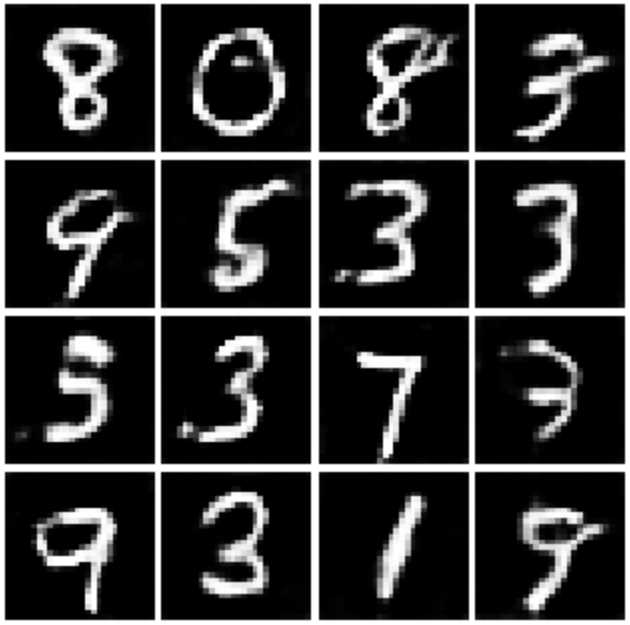
Iter: 500



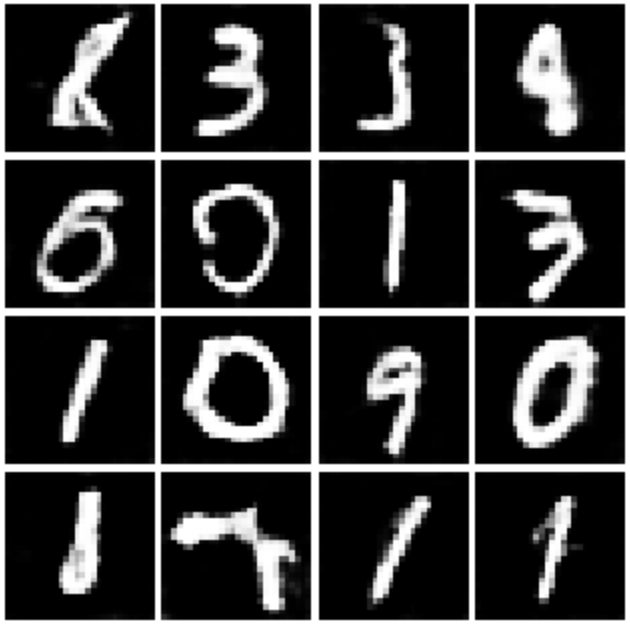
Iter: 750



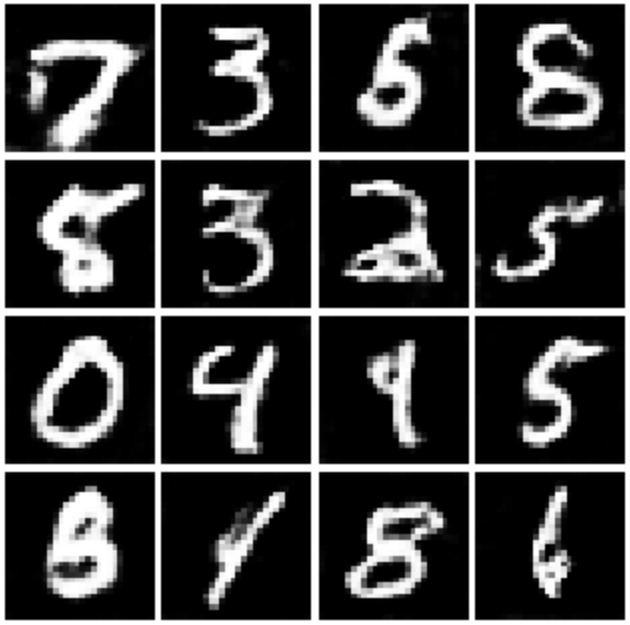
Iter: 1000



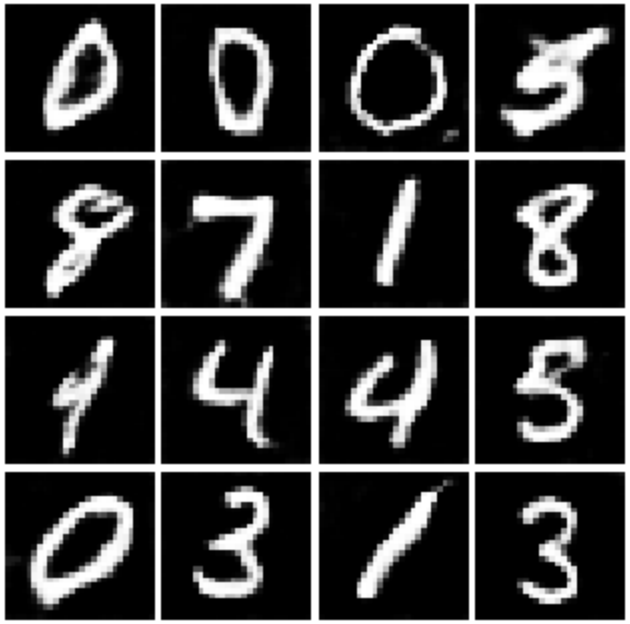
Iter: 1250



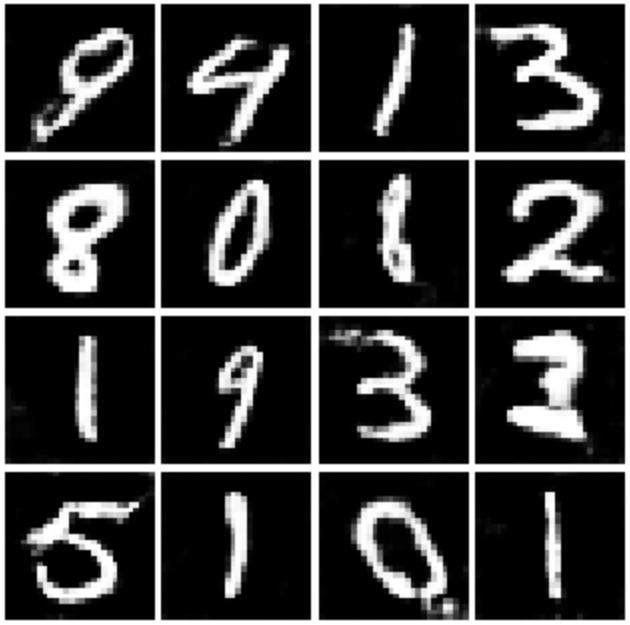
Iter: 1500



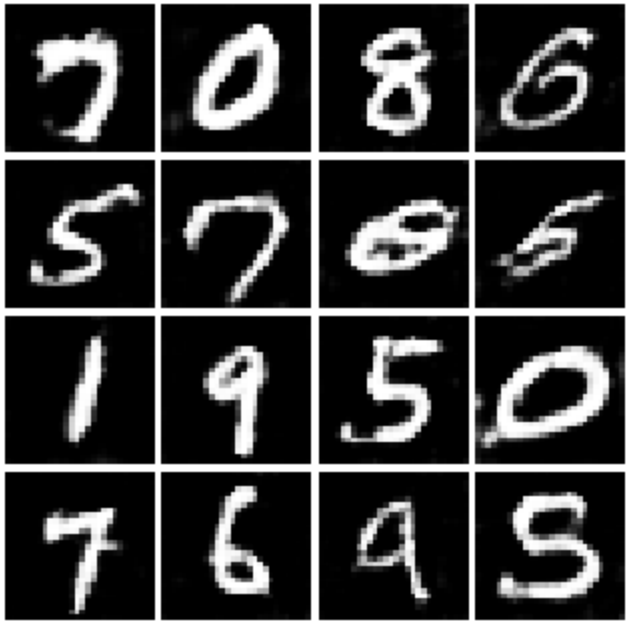
Iter: 1750



Iter: 2000



Iter: 2250

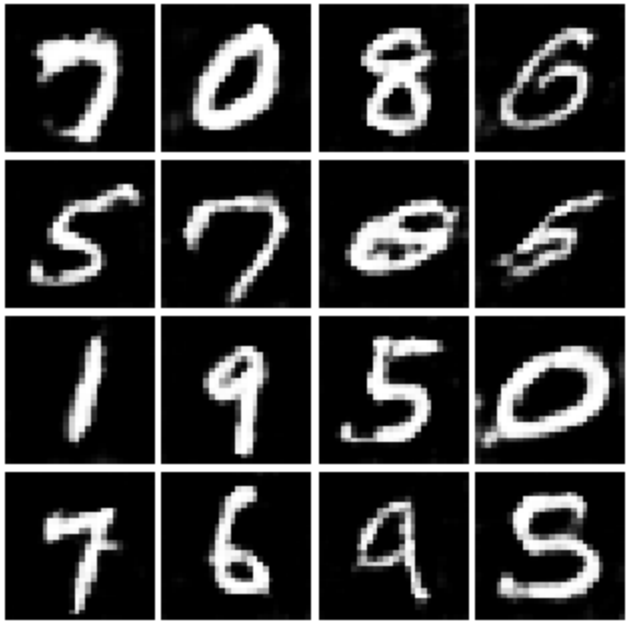


Inline Question 3

What does your final DCGAN image look like?

In [23]:
```
# This output is your answer.
print('DCGAN final image:')
show_images(images[-1])
plt.show()
```

```
DCGAN final image:
```



Inline Question 4 (1 point)

We will look at an example to see why alternating minimization of the same objective (like in a GAN) can be tricky business.

Consider $f(x, y) = xy$. What does $\min_x \max_y f(x, y)$ evaluate to? (0.4 points). Hint: minmax tries to minimize the maximum value achievable. Hint: Consider 3 separate cases: $x > 0$, $x = 0$, and $x < 0$.

Now try to evaluate this function numerically for 6 steps, starting at the point $(1, 1)$, by using alternating gradient (first updating y, then updating x using that updated y) with step size 1. **Here step size is the learning_rate, and steps will be learning_rate * gradient.** You'll find that writing out the update step in terms of $x_t, y_t, x_{t+1}, y_{t+1}$ will be useful.

Record the six pairs of explicit values for (x_t, y_t) in the table below and briefly explain what $f(x, y)$ evaluates to (0.6 points).

i.e., for $\eta = 1$, calculate:

For $i = 1, 2, 3, \ldots$

1. $y_i = y_{i-1} + \eta \frac{\partial f(x,y)}{\partial y}|_{x=x_{i-1}, y=y_{i-1}}$

2. $x_i = x_{i-1} - \eta \frac{\partial f(x,y)}{\partial x}|_{x=x_{i-1},y=y_i}$

Your answer:

$\min_x \max_y f(x,y) = 0$ since when $x > 0$, $\max_y f(x,y) = +\infty$, when $x = 0$, $\max_y f(x,y) = 0$ and when $x < 0$, $\max_y f(x,y) = +\infty$. So the minimum is 0.

y_0	y_1	y_2	y_3	y_4	y_5	y_6
1	2	1	-1	-2	-1	1
x_0	x_1	x_2	x_3	x_4	x_5	x_6
1	-1	-2	-1	1	2	1

Inline Question 5 (1 point)

Using this method, will we ever reach the optimal value? Why or why not?

Your answer:

No. Because y_0 and x_0 would go back to the original value after 6 iterations. And the cycle would repeat.

Inline Question 6 (1 point)

If the generator loss decreases during training while the discriminator loss stays at a constant high value from the start, is this a good sign? Why or why not? A qualitative answer is sufficient.

Your answer:

Possibly not a good sign. There could be training imbalance or a very weak discriminator.

VAE/vae.py

```python
import torch
import torch.nn as nn
from torch import Tensor
from torch.nn import functional as F


class VAE(nn.Module):
    def __init__(self, input_size: int, latent_size: int = 15):
        super().__init__()
        self.input_size = input_size  # H*W
        self.latent_size = latent_size  # Z
        self.hidden_dim = None  # H_d
        self.encoder = None
        self.mu_layer = None
        self.logvar_layer = None
        self.decoder = None


        ############################################################################################
        # TODO: Implement the encoder and decoder architectures described in the notebook.
        #
        #
        # Encoder: self.encoder maps (N, 1, H, W) → (N, H_d) via nn.Sequential.
        #
        #   Then self.mu_layer and self.logvar_layer each map (N, H_d) → (N, Z).
        #
        #   Set self.hidden_dim = 400.
        #
        #
        # Decoder: self.decoder maps (N, Z) → (N, 1, H, W) via nn.Sequential.
        #
        #   Use nn.Unflatten to reshape the output back to image dimensions.
        #

        ############################################################################################
        # *****START OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****

        self.hidden_dim = 400
        self.encoder = nn.Sequential(
            nn.Flatten(),
            nn.Linear(self.input_size, self.hidden_dim),
            nn.ReLU(),
            nn.Linear(self.hidden_dim, self.hidden_dim),
            nn.ReLU(),
            nn.Linear(self.hidden_dim, self.hidden_dim),
            nn.ReLU()
        )
```

```
40
41          self.mu_layer = nn.Linear(self.hidden_dim, self.latent_size)
42          self.logvar_layer = nn.Linear(self.hidden_dim, self.latent_size)
43
44          self.decoder = nn.Sequential(
45              nn.Linear(self.latent_size, self.hidden_dim),
46              nn.ReLU(),
47              nn.Linear(self.hidden_dim, self.hidden_dim),
48              nn.ReLU(),
49              nn.Linear(self.hidden_dim, self.hidden_dim),
50              nn.ReLU(),
51              nn.Linear(self.hidden_dim, self.input_size),
52              nn.Sigmoid(),
53              nn.Unflatten(1, (1, 28, 28))
54          )
55
56          # *****END OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
57
##############################################################################################
58          #                                       END OF YOUR CODE
#
59
##############################################################################################
60
61      def forward(self, x: Tensor) -> tuple[Tensor, Tensor, Tensor]:
62          """
63          Performs forward pass through FC-VAE model by passing image through
64          encoder, reparametrize trick, and decoder models.
65
66          Inputs:
67          - x: Batch of input images of shape (N, 1, H, W)
68
69          Returns:
70          - x_hat: Reconstructed input data of shape (N, 1, H, W)
71          - mu: Estimated posterior mu of shape (N, Z)
72          - logvar: Estimated variance in log-space of shape (N, Z)
73          """
74          x_hat = None
75          mu = None
76          logvar = None
77
##############################################################################################
78          # TODO: Implement the forward pass:
#
79          # (1) Pass x through self.encoder, then self.mu_layer and self.logvar_layer
#
80          # (2) Call reparametrize(mu, logvar) to get the latent vector z
#
81          # (3) Pass z through self.decoder to reconstruct x
#
```

```
82
    ##########################################################################################
83          # *****START OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
84
85          x = self.encoder(x)
86          mu = self.mu_layer(x)
87          logvar = self.logvar_layer(x)
88          z = reparametrize(mu, logvar)
89          x_hat = self.decoder(z)
90
91          # *****END OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
92
    ##########################################################################################
93          #                                           END OF YOUR CODE
    #
94
    ##########################################################################################
95          return x_hat, mu, logvar
96
97
98  class CVAE(nn.Module):
99      def __init__(self, input_size: int, num_classes: int = 10, latent_size: int = 15):
100         super().__init__()
101         self.input_size = input_size  # H*W
102         self.latent_size = latent_size  # Z
103         self.num_classes = num_classes  # K
104         self.hidden_dim = None  # H_d
105         self.encoder = None
106         self.mu_layer = None
107         self.logvar_layer = None
108         self.decoder = None
109
110
    ##########################################################################################
111         # TODO: Same architecture as VAE, but:
    #
112         # - Encoder input size is (H*W + K): the flattened image concatenated with one-hot
    label  #
113         # - Decoder input size is (Z + K): the latent vector concatenated with one-hot label
    #
114         # The flatten + cat operations happen in forward(), not here.
    #
115
    ##########################################################################################
116         # *****START OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
117
118         self.hidden_dim = 400
119         side = int(self.input_size**0.5)
120
121         self.encoder = nn.Sequential(
122             nn.Linear(self.input_size + self.num_classes, self.hidden_dim),
```

```
            nn.ReLU(),
        )

        self.mu_layer = nn.Linear(self.hidden_dim, self.latent_size)
        self.logvar_layer = nn.Linear(self.hidden_dim, self.latent_size)

        self.decoder = nn.Sequential(
            nn.Linear(self.latent_size + self.num_classes, self.hidden_dim),
            nn.ReLU(),
            nn.Linear(self.hidden_dim, self.input_size),
            nn.Sigmoid(),
            nn.Unflatten(1, (1, side, side)),
        )

        # *****END OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****

################################################################################################
        #                                          END OF YOUR CODE
#
################################################################################################

    def forward(self, x: Tensor, labels: Tensor) -> tuple[Tensor, Tensor, Tensor]:
        """
        Performs forward pass through FC-CVAE model.

        Inputs:
        - x: Input data of shape (N, 1, H, W)
        - labels: One-hot class vector of shape (N, K)

        Returns:
        - x_hat: Reconstructed input data of shape (N, 1, H, W)
        - mu: Estimated posterior mu of shape (N, Z)
        - logvar: Estimated variance in log-space of shape (N, Z)
        """
        x_hat = None
        mu = None
        logvar = None

################################################################################################
        # TODO: Implement the forward pass:
#
        # (1) Flatten x and cat with labels, pass through encoder → mu, logvar
#
        # (2) Reparametrize to get z
#
        # (3) Cat z with labels, pass through decoder → x_hat
#

################################################################################################
        # *****START OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
```

```
        x_flattened = x.view(x.size(0), -1)
        encoder_input = torch.cat((x_flattened, labels), dim=1)

        hidden = self.encoder(encoder_input)

        mu = self.mu_layer(hidden)
        logvar = self.logvar_layer(hidden)

        z = reparametrize(mu, logvar)
        decoder_input = torch.cat((z, labels), dim=1)

        x_hat = self.decoder(decoder_input)

        # *****END OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****

##############################################################################################
        #                                          END OF YOUR CODE
#
##############################################################################################
        return x_hat, mu, logvar


def reparametrize(mu: Tensor, logvar: Tensor) -> Tensor:
    """
    Differentiably sample random Gaussian data with specified mean and variance
    using the reparameterization trick.

    We sample epsilon ~ N(0, 1) and return z = mu + sigma * epsilon, where
    sigma = exp(0.5 * logvar). This lets gradients flow through z to mu and logvar.

    Inputs:
    - mu: Tensor of shape (N, Z) giving means
    - logvar: Tensor of shape (N, Z) giving log-variances

    Returns:
    - z: Tensor of shape (N, Z), each z[i,j] ~ N(mu[i,j], exp(logvar[i,j]))

    HINT: Use torch.randn_like(mu) to sample epsilon — it automatically matches
    the device and dtype of mu so you don't need to move anything to GPU manually.
    """
    z = None
    # *****START OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****

    epsilon = torch.randn_like(mu)
    sigma = torch.exp(0.5 * logvar)
    z = mu + sigma * epsilon

    # *****END OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
    return z
```

```python


def loss_function(x_hat: Tensor, x: Tensor, mu: Tensor, logvar: Tensor) -> Tensor:
    """
    Computes the negative variational lower bound loss term of the VAE.

    Loss = reconstruction_loss + KL_divergence, averaged over the batch.

    Reconstruction loss: binary cross entropy between x_hat and x.
    KL divergence (closed form for diagonal Gaussian vs standard normal):
        -0.5 * sum(1 + logvar - mu^2 - exp(logvar))

    Inputs:
    - x_hat: Reconstructed input data of shape (N, 1, H, W)
    - x: Original input data of shape (N, 1, H, W)
    - mu: Posterior mean of shape (N, Z)
    - logvar: Posterior log-variance of shape (N, Z)

    Returns:
    - loss: Scalar tensor containing the mean loss over the batch.

    HINT: Use F.binary_cross_entropy(x_hat, x, reduction='sum') for the
    reconstruction term, then divide the total loss by batch size (mu.shape[0]).
    """
    loss = None
    # *****START OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****

    batch_size = mu.shape[0]

    reconstruction_loss = F.binary_cross_entropy(x_hat, x, reduction='sum')

    kl_divergence = -0.5 * torch.sum(1 + logvar - mu.pow(2) - logvar.exp())

    loss = (reconstruction_loss + kl_divergence) / batch_size


    # *****END OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
    return loss

```

We would like to acknowledge University of Michigan's EECS 498-007/598-005 on which we based the development of this project.

Variational Autoencoder

In this notebook, you will implement a variational autoencoder (VAE) and a conditional variational autoencoder (CVAE) and apply them to the MNIST handwritten digit dataset. An autoencoder learns a compact latent representation of training images and reconstructs them from that representation. The VAE extends this by making the encoder and decoder probabilistic, allowing us to sample from the learned latent distribution to generate new images.

Assignment Overview

All your code goes in `vae.py`. The notebook imports from it and runs checks — do not modify the notebook cells.

Task	Points
`VAE.__init__` — encoder, `mu_layer`, `logvar_layer`	4
`VAE.__init__` — decoder	1
`VAE.forward`	1
`reparametrize`	2
`loss_function`	1
`CVAE.__init__` + `CVAE.forward`	3
Total	**12**

`train_vae`, `show_images`, `reset_seed`, `rel_error`, and `one_hot` are all provided in `vae_utils.py` — do not modify that file.

In [1]:
```
import os
import sys
```

```
sys.path.insert(0, os.path.dirname(os.path.abspath('vae.py')))
print('cwd:', os.getcwd())
```

```
cwd: /content
```

In [1]:

In [2]:
```
import importlib
import sys
sys.modules["imp"] = importlib

%load_ext autoreload
%autoreload 2

import torch
import torch.nn as nn
import torch.nn.functional as F
import torch.optim as optim
import torchvision.transforms as T
import torchvision.datasets as dset
from torch.utils.data import DataLoader
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec

from vae_utils import rel_error, reset_seed, show_images, train_vae, count_params

%matplotlib inline
plt.rcParams['figure.figsize'] = (10.0, 8.0)
plt.rcParams['font.size'] = 16
plt.rcParams['image.interpolation'] = 'nearest'
plt.rcParams['image.cmap'] = 'gray'

device = (
    torch.device('cuda') if torch.cuda.is_available()
    else torch.device('mps') if torch.backends.mps.is_available()
    else torch.device('cpu')
)
print('Using device:', device)
```

```
Using device: cuda
```

Load MNIST Dataset

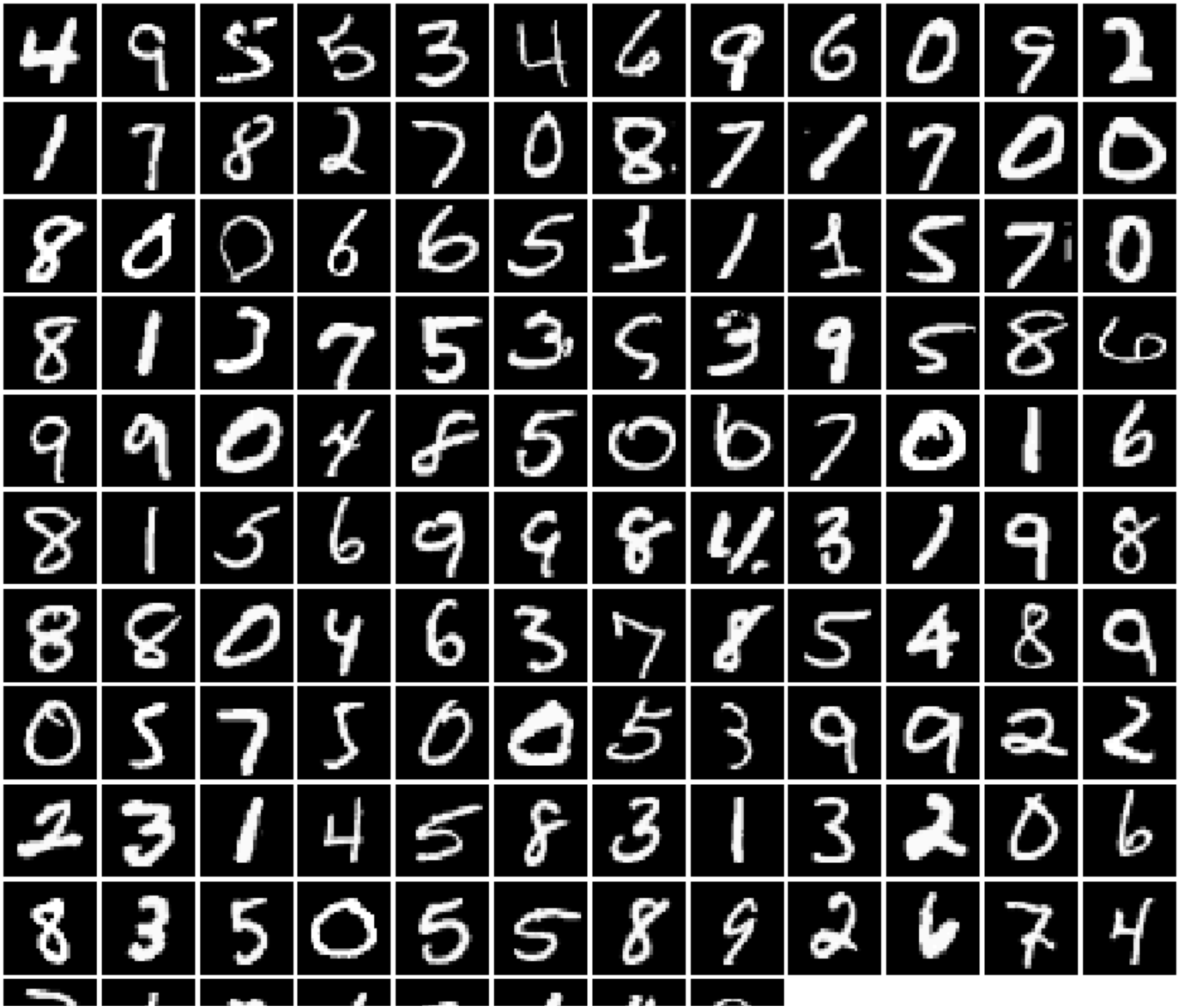
VAEs are notoriously finicky with hyperparameters, and also require many training epochs. In order to make this assignment approachable, we will be working on the MNIST dataset, which is 60,000 training and 10,000 test images. Each picture contains a centered image of white digit on black background (0 through 9). The data will be saved into a folder called `MNIST` on first run.

In [3]:
```
batch_size = 128

mnist_train = dset.MNIST('./mnist_data', train=True, download=True, transform=T.ToTensor())
loader_train = DataLoader(mnist_train, batch_size=batch_size, shuffle=True, drop_last=True, num_workers=0)
```

```
100%|██████████| 9.91M/9.91M [00:00<00:00, 19.4MB/s]
100%|██████████| 28.9k/28.9k [00:00<00:00, 488kB/s]
100%|██████████| 1.65M/1.65M [00:00<00:00, 4.70MB/s]
100%|██████████| 4.54k/4.54k [00:00<00:00, 32.7MB/s]
```

Visualize Dataset

In [4]:
```
imgs = next(iter(loader_train))[0]
show_images(imgs)
```



Fully Connected VAE

Our first VAE implementation will consist solely of fully connected layers. We'll take the `1 x 28 x 28` shape of our input and flatten the features to create an input dimension size of 784. In this section you'll define the encoder and decoder in the `VAE` class in `vae.py`, implement the reparametrization trick, the forward pass, and the loss function.

FC-VAE Encoder (4 points)

Build the encoder inside `VAE.__init__` in `vae.py`. Set `self.hidden_dim = 400` and define:

- `self.encoder` — an `nn.Sequential` that maps `(N, 1, H, W)` → `(N, hidden_dim)`
- `self.mu_layer` — a single `nn.Linear` that maps `(N, hidden_dim)` → `(N, Z)`
- `self.logvar_layer` — a single `nn.Linear` that maps `(N, hidden_dim)` → `(N, Z)`

Encoder architecture:

- `nn.Flatten()`
- Linear(`input_size` → `hidden_dim`) + ReLU
- Linear(`hidden_dim` → `hidden_dim`) + ReLU
- Linear(`hidden_dim` → `hidden_dim`) + ReLU

In [5]:
```
from vae import VAE

def test_encoder(model, input_size, hidden_dim, n_encoder_lin_layers):
    expected = (input_size + 1) * hidden_dim + (n_encoder_lin_layers - 1) * (hidden_dim + 1) * hidden_dim
    actual = count_params(model.encoder)
    if actual == expected:
        print('Correct number of parameters in model.encoder.')
    else:
        print('Incorrect number of parameters in model.encoder. (mu_layer and logvar_layer should be separate)')
```

```
def test_mu_logvar(model, hidden_dim, latent_size):
    for name, layer in [('mu_layer', model.mu_layer), ('logvar_layer', model.logvar_layer)]:
        expected = (hidden_dim + 1) * latent_size
        actual = count_params(layer)
        status = 'Correct' if actual == expected else 'Incorrect'
        print(f'{status} number of parameters in model.{name}.')

test_encoder(VAE(345, 17), 345, 400, 3)
test_mu_logvar(VAE(345, 17), 400, 17)
```

```
Correct number of parameters in model.encoder.
Correct number of parameters in model.mu_layer.
Correct number of parameters in model.logvar_layer.
```

FC-VAE Decoder (1 point)

Build the decoder inside `VAE.__init__`. Define `self.decoder` as an `nn.Sequential` that maps `(N, Z)` → `(N, 1, H, W)`.

Decoder architecture:

- Linear(`latent_size` → `hidden_dim`) + ReLU
- Linear(`hidden_dim` → `hidden_dim`) + ReLU
- Linear(`hidden_dim` → `hidden_dim`) + ReLU
- Linear(`hidden_dim` → `input_size`) + Sigmoid
- `nn.Unflatten(1, (1, 28, 28))`

In [6]:
```
from vae import VAE

def test_decoder(model, input_size, hidden_dim, latent_size, n_decoder_lin_layers):
    expected = (
        (latent_size + 1) * hidden_dim
        + (n_decoder_lin_layers - 2) * (hidden_dim + 1) * hidden_dim
        + (hidden_dim + 1) * input_size
    )
    actual = count_params(model.decoder)
    status = 'Correct' if actual == expected else 'Incorrect'
    print(f'{status} number of parameters in model.decoder.')

test_decoder(VAE(345, 17), 345, 400, 17, 4)
```

```
Correct number of parameters in model.decoder.
```

Reparametrization (2 points)

To sample a latent vector z from $q_\phi(z|x) = \mathcal{N}(\mu, \sigma^2)$ in a differentiable way, we use the reparametrization trick:

$$z = \mu + \sigma \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, 1)$$

This lets gradients flow back through z to μ and σ during backpropagation.

Implement `reparametrize` in `vae.py`. Verify that your mean and std errors are below `1e-4`.

In [7]:
```
reset_seed(0)
from vae import reparametrize

latent_size = 15
size = (1, latent_size)
mu = torch.zeros(size)
logvar = torch.ones(size)

z = reparametrize(mu, logvar)

expected_mean = torch.FloatTensor([-0.4363])
expected_std = torch.FloatTensor([1.6860])
assert z.size() == size
print('Mean Error', rel_error(torch.mean(z, dim=-1), expected_mean))
print('Std Error',  rel_error(torch.std(z, dim=-1),  expected_std))
```

```
Mean Error 5.639056398351415e-05
Std Error 7.1412955526273885e-06
```

FC-VAE Forward (1 point)

Implement `VAE.forward` in `vae.py`. Pass the input through the encoder to get `mu` and `logvar`, reparametrize to get `z`, then decode `z` to reconstruct the image.

In [8]:
```
from vae import VAE
```

```python
def test_VAE_shapes():
    all_correct = True
    with torch.no_grad():
        batch_size = 3
        latent_size = 17
        x_hat, mu, logvar = VAE(28 * 28, latent_size)(torch.ones(batch_size, 1, 28, 28))
        for name, tensor, expected in [
            ('x_hat',  x_hat,  (batch_size, 1, 28, 28)),
            ('mu',     mu,     (batch_size, latent_size)),
            ('logvar', logvar, (batch_size, latent_size)),
        ]:
            if tuple(tensor.shape) != expected:
                print(f'{name}: expected {expected}, got {tuple(tensor.shape)}')
                all_correct = False
        if all_correct:
            print('Shapes of x_hat, mu, and logvar are correct.')
        if batch_size > 1 and x_hat.std(0).mean() == 0:
            print('Warning: x_hat has no randomness — check reparametrize.')

test_VAE_shapes()
```

```
Shapes of x_hat, mu, and logvar are correct.
```

Loss Function (1 point)

The VAE loss is the negative variational lower bound:

$$\mathcal{L} = \underbrace{-\mathbb{E}_{z\sim q_\phi}[\log p_\theta(x|z)]}_{\text{reconstruction}} + \underbrace{D_{KL}(q_\phi(z|x)\|p(z))}_{\text{KL divergence}}$$

The KL term has a closed form for diagonal Gaussians vs. a standard normal prior:

$$D_{KL} = -\frac{1}{2}\sum_{j=1}^{Z}\left(1 + \log\sigma_j^2 - \mu_j^2 - \sigma_j^2\right)$$

Implement `loss_function` in `vae.py`. Use `F.binary_cross_entropy(x_hat, x, reduction='sum')` for the reconstruction term. Average the total loss over the batch. Your relative error should be ≤ `1e-5`.

In [9]:
```
from vae import loss_function

image_hat = torch.sigmoid(torch.FloatTensor([[2, 5], [6, 7]]).unsqueeze(0).unsqueeze(0))
image     = torch.sigmoid(torch.FloatTensor([[1, 10], [9, 3]]).unsqueeze(0).unsqueeze(0))
image_hat = torch.tile(image_hat, (10, 1, 1, 1))
image     = torch.tile(image, (10, 1, 1, 1))

mu_shape = (image.shape[0], 15)
mu, logvar = torch.ones(mu_shape), torch.zeros(mu_shape)

expected_out = torch.tensor(1.0079 + 7.5000)
out = loss_function(image_hat, image, mu, logvar)
print('Loss error', rel_error(expected_out, out))
```

```
Loss error 2.1297676389877955e-06
```

Train VAE

`train_vae` is provided in `vae_utils.py`. Training for 10 epochs should take ~2 minutes and your loss should be less than 120.

In [10]:
```
from vae import VAE

num_epochs = 10
latent_size = 15
input_size = 28 * 28

vae_model = VAE(input_size, latent_size=latent_size).to(device)
optimizer = optim.Adam(vae_model.parameters(), lr=1e-3)

for epoch in range(num_epochs):
    train_vae(epoch, vae_model, loader_train, optimizer)
```

```
Epoch 0: loss = 190.0974
Epoch 1: loss = 143.5985
Epoch 2: loss = 130.7988
Epoch 3: loss = 126.8050
Epoch 4: loss = 124.0375
Epoch 5: loss = 121.0722
Epoch 6: loss = 118.3619
Epoch 7: loss = 116.4784
Epoch 8: loss = 115.0019
Epoch 9: loss = 113.9125
```

Visualize Results

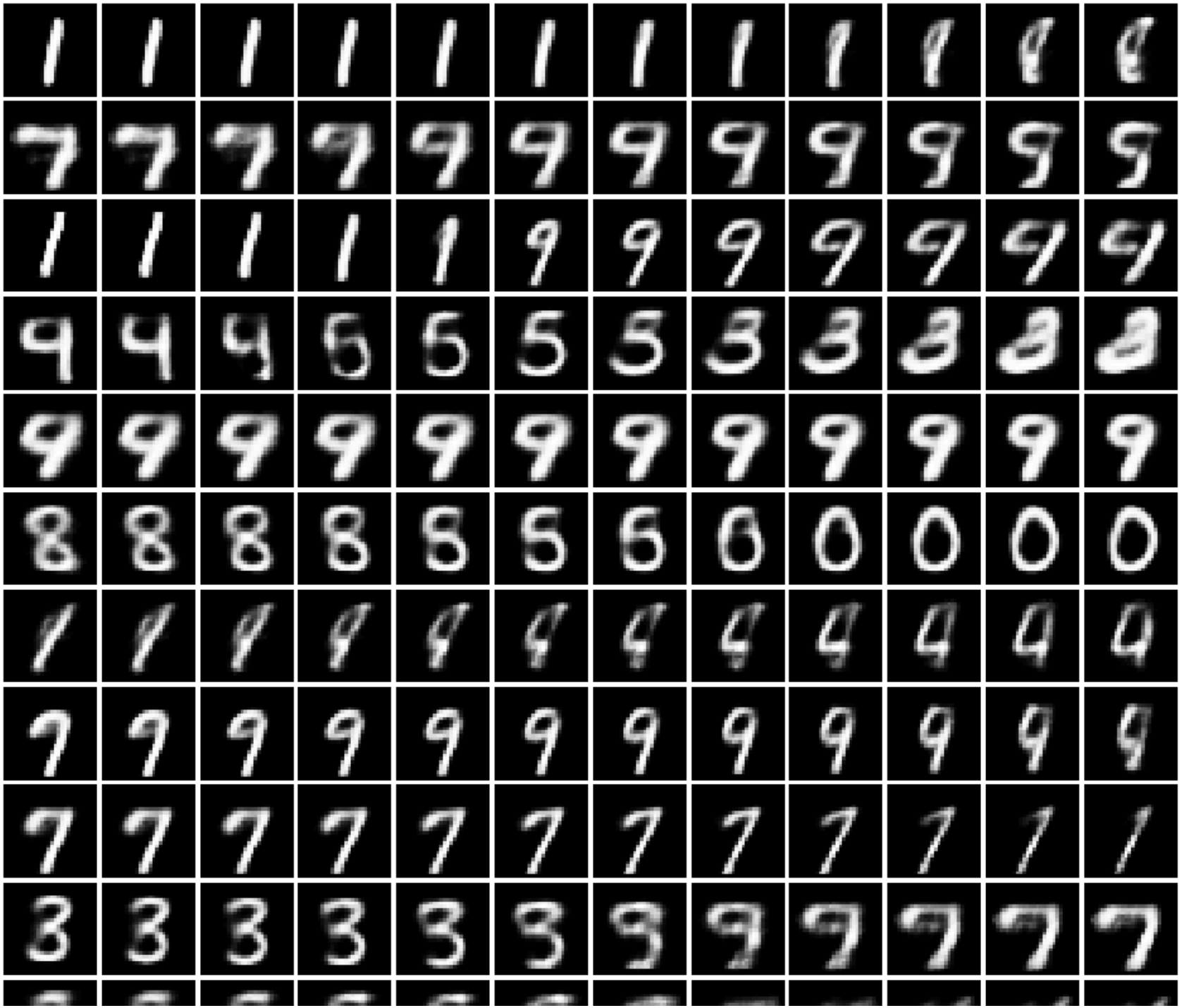
Sample random latent vectors and decode them to generate new images. You should be able to visually recognize many digits, though some may be blurry.

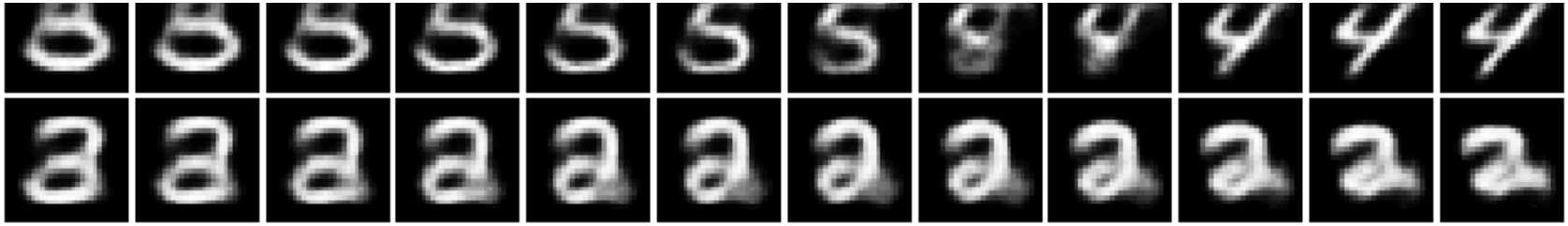
In [11]:
```
vae_model.eval()
with torch.no_grad():
    z = torch.randn(10, latent_size, device=device)
    samples = vae_model.decoder(z).cpu()

fig = plt.figure(figsize=(10, 1))
gspec = gridspec.GridSpec(1, 10)
gspec.update(wspace=0.05, hspace=0.05)
for i, sample in enumerate(samples):
    ax = plt.subplot(gspec[i])
    plt.axis('off')
    ax.set_aspect('equal')
    plt.imshow(sample.reshape(28, 28), cmap='Greys_r')
```

Latent Space Interpolation

Generate random latent vectors z_0 and z_1 and linearly interpolate between them. Each row below interpolates between two random vectors — you should see smooth transitions between digits.

In [12]:
```
S = 12
vae_model.eval()
with torch.no_grad():
    z0 = torch.randn(S, latent_size, device=device)
    z1 = torch.randn(S, latent_size, device=device)
    w  = torch.linspace(0, 1, S, device=device).view(S, 1, 1)
    z  = (w * z0 + (1 - w) * z1).transpose(0, 1).reshape(S * S, latent_size)
    x  = vae_model.decoder(z).cpu()
show_images(x)
```





Conditional FC-VAE

The CVAE extends the VAE by conditioning on the class label c. Instead of $q_\phi(z|x)$ and $p_\theta(x|z)$ we have $q_\phi(z|x,c)$ and $p_\theta(x|z,c)$. This lets us generate specific digits at inference time by choosing c.

CVAE (3 points)

Implement `CVAE.__init__` and `CVAE.forward` in `vae.py`. Use the same architecture as the VAE with these modifications:

1. **Encoder** input: concatenation of flattened image `(N, H*W)` and one-hot label `(N, K)` → first linear layer takes `H*W + K`. Do **not** put `nn.Flatten` inside the encoder Sequential — flatten in `forward` instead.
2. **Decoder** input: concatenation of latent vector `z` `(N, Z)` and one-hot label `(N, K)` → first linear layer takes `Z + K`.
3. **Forward**: flatten `x`, `torch.cat` with `labels` before encoder; `torch.cat` `z` with `labels` before decoder.

In [13]:
```
from vae import CVAE

def test_CVAE_shapes():
    all_correct = True
    with torch.no_grad():
        batch_size = 3
        num_classes = 10
        latent_size = 17
        cls = F.one_hot(torch.tensor([3] * batch_size), num_classes=num_classes).float()
        x_hat, mu, logvar = CVAE(28 * 28, num_classes=num_classes, latent_size=latent_size)(
            torch.ones(batch_size, 1, 28, 28), cls
        )
        for name, tensor, expected in [
            ('x_hat',  x_hat,  (batch_size, 1, 28, 28)),
```

```
            ('mu',     mu,     (batch_size, latent_size)),
            ('logvar', logvar, (batch_size, latent_size)),
        ]:
            if tuple(tensor.shape) != expected:
                print(f'{name}: expected {expected}, got {tuple(tensor.shape)}')
                all_correct = False
        if all_correct:
            print('Shapes of x_hat, mu, and logvar are correct.')
        if batch_size > 1 and x_hat.std(0).mean() == 0:
            print('Warning: x_hat has no randomness.')

test_CVAE_shapes()
```

```
Shapes of x_hat, mu, and logvar are correct.
```

Train CVAE

Training for 10 epochs should take ~2 minutes and your loss should be less than 120.

In [14]:
```
from vae import CVAE

num_epochs = 10
latent_size = 15
input_size = 28 * 28

cvae = CVAE(input_size, latent_size=latent_size).to(device)
optimizer = optim.Adam(cvae.parameters(), lr=1e-3)

for epoch in range(num_epochs):
    train_vae(epoch, cvae, loader_train, optimizer, cond=True)
```

```
Epoch 0: loss = 162.1331
Epoch 1: loss = 119.6202
Epoch 2: loss = 113.1262
Epoch 3: loss = 109.8344
Epoch 4: loss = 107.7104
Epoch 5: loss = 106.2335
Epoch 6: loss = 105.1673
Epoch 7: loss = 104.2920
Epoch 8: loss = 103.5530
Epoch 9: loss = 102.9455
```

Visualize CVAE Results

We can now conditionally generate one sample per digit (0–9) by providing the corresponding one-hot label. Each digit should be reasonably recognizable.

In [25]:
```
cvae.eval()
with torch.no_grad():
    z = torch.randn(10, latent_size)
    c = torch.eye(10, 10)  # one-hot labels for digits 0-9
    zc = torch.cat((z, c), dim=-1).to(device)
    samples = cvae.decoder(zc).cpu()

fig = plt.figure(figsize=(10, 1))
gspec = gridspec.GridSpec(1, 10)
gspec.update(wspace=0.05, hspace=0.05)
for i, sample in enumerate(samples):
    ax = plt.subplot(gspec[i])
    plt.axis('off')
    ax.set_aspect('equal')
    plt.imshow(sample.reshape(28, 28), cmap='Greys_r')
```